

# 1 When do single-cell response distributions improve drug 2 retrieval?

## 4 Abstract

5 Artificial intelligence virtual cells aim to predict cellular responses to perturbations and, ultimately,  
6 to guide intervention selection. Conventional transcriptomic drug retrieval represents each pertur-  
7 bation by a single expression signature, whereas single-cell perturbation atlases preserve cell-to-cell  
8 variation in perturbation responses. Here we define population-to-population drug retrieval (PopRe-  
9 trieve), in which candidate drugs are ranked by comparing single-cell response populations, and ask  
10 when this additional information improves drug ranking. We find that most of the gain over direction-  
11 only mean matching is recovered by a mean-based score that also retains response magnitude, with  
12 a smaller additional gain from retaining the full response population. The advantage of population-  
13 level retrieval weakens as evaluation moves away from response matching: it disappears or reverses  
14 under mechanism-of-action recovery, and most of its association with an external functional readout is  
15 reproduced by response magnitude alone. The same held-out protein measurements favour different  
16 retrieval strategies depending on whether they are summarized as means or retained as populations,  
17 showing that the statistical form of an external evaluator can influence which retrieval strategy ap-  
18 pears preferable. With candidate responses observed, population-level retrieval corrects an incorrect  
19 top-ranked choice about four times as often as it overturns a correct one relative to magnitude-aware  
20 mean matching, and about five times relative to direction-only matching, but the gain is confined to  
21 the roughly one-fifth of queries for which mean matching leaves room for improvement. When candi-  
22 date responses are predicted rather than observed, this oracle advantage disappears and can reverse  
23 sign even when mean-level retrieval performance is preserved. Together, these results show that  
24 the value of single-cell population information depends on whether information beyond the mean  
25 changes candidate ordering and, when required, survives forward prediction.  
26

27 Keywords: artificial intelligence virtual cell; single-cell perturbation; drug retrieval; population-to-  
28 population retrieval; cellular heterogeneity; objective–utility mismatch

## 30 Introduction

31 Artificial intelligence virtual cells<sup>1,2</sup> aim to predict cellular responses to perturbations<sup>3–5</sup> and, ultimately,  
32 to help prioritize interventions for testing. These goals pose two related problems. Forward prediction  
33 asks what response a perturbation will produce<sup>6–9</sup>; retrieval asks which perturbation best matches a  
34 cellular response of interest<sup>10</sup>. Transcriptomic drug retrieval has traditionally represented each drug by  
35 a single perturbation signature and ranked candidates by similarity or reversal<sup>11,12</sup>. This approach has  
36 made large drug libraries searchable, but collapses variation across cells into a single summary. Single-  
37 cell perturbation atlases now capture this variation<sup>13–15</sup>, including state-dependent responses<sup>16</sup> and ef-  
38 fects confined to minority cell populations<sup>17</sup>. Here we define population-to-population drug retrieval  
39 (PopRetrieve), in which candidate drugs are ranked by comparing single-cell response populations, and  
40 ask when this additional information improves drug ranking.

41 More detailed response information does not necessarily lead to a better drug ranking. Two drugs  
42 can have similar average responses yet differ in how individual cells respond, creating information that  
43 a population-aware method can use. But an apparent advantage over classical signature matching can  
44 arise for more than one reason. Population comparison retains response magnitude that directional sim-  
45 ilarity discards, so a gain over direction-only matching need not arise from within-population structure.  
46 The apparent benefit also depends on how retrieval quality is evaluated: a criterion that preserves full  
47 response distributions<sup>18,19</sup> will naturally favour information that a population-aware scorer retains<sup>20-22</sup>.  
48 Such gains establish better response matching, but not necessarily a better biological or functional rank-  
49 ing<sup>23-27</sup>.

50 We therefore decompose the problem into information retained, information that changes candi-  
51 date ordering, and information that survives prediction. We first distinguish three scoring rules by how  
52 much of the response each retains: direction, direction plus magnitude, and the full response population.  
53 This separates gains recovered by restoring response magnitude from those that require information be-  
54 yond the mean. We then evaluate the resulting rankings using increasingly task-distal criteria: response  
55 matching, mechanism recovery and external functional readouts<sup>28,29</sup>. We use state-dependent pertur-  
56 bation responses as a concrete mechanism through which population information can affect retrieval.  
57 For such information to matter, it must be present and recoverable, change candidate ordering beyond  
58 what the mean already determines and, when candidate responses are not observed, survive forward  
59 prediction. This separates the existence of single-cell response structure from its value for intervention  
60 selection.

61 Here we show that most of the advantage of population comparison over direction-only signature  
62 matching is recovered by a mean-based score that also retains response magnitude, leaving a smaller  
63 gain that requires the full response population. The advantage of population comparison weakens as  
64 evaluation moves away from response matching: it disappears or reverses under mechanism-of-action  
65 recovery, and most of its association with an external functional readout is reproduced by response mag-  
66 nitude alone. The same held-out protein measurements favour different retrieval strategies depending  
67 only on whether they are summarized as means or retained as populations, showing that external eval-  
68 uation is not automatically neutral. In a source-to-target intervention-identification task, population  
69 information provides a smaller additional gain beyond magnitude-aware mean matching when candi-  
70 date responses are observed, and its whole advantage over direction-only mean matching is confined  
71 to the queries for which that baseline leaves room for improvement. That advantage does not survive  
72 when candidate responses must instead be predicted. Together, these results show that the value of  
73 single-cell population information depends not simply on whether additional cellular structure exists,  
74 but on whether it changes candidate ordering and survives to the final retrieval decision.

## 75 Results

### 76 Population-to-population retrieval generalizes classical signature matching

77 Population-to-population retrieval ranks candidates by comparing query and candidate response popu-  
78 lations directly (Fig. 1a,b), whereas classical mean-signature retrieval<sup>11,12</sup> first collapses each population  
79 to a single mean differential-expression vector. Retaining the cells makes response structure available  
80 that averaging removes, including differences between cellular states or subpopulations. The ranking  
81 task is the same; the representation determines how much measured response reaches the decision.

82 Two candidate populations can share the same mean response and still differ markedly in how indi-  
83 vidual cells respond (Fig. 1c,d). A population-aware score can separate such candidates because it retains

84 distributional structure that any mean-only score discards. Whether that additional information matters  
 85 for retrieval is a decision question: it matters only when it changes relative candidate scores enough to  
 86 alter candidate ordering (Fig. 1e). This distinction between information retained and information that  
 87 changes the ranking motivates the analyses that follow.

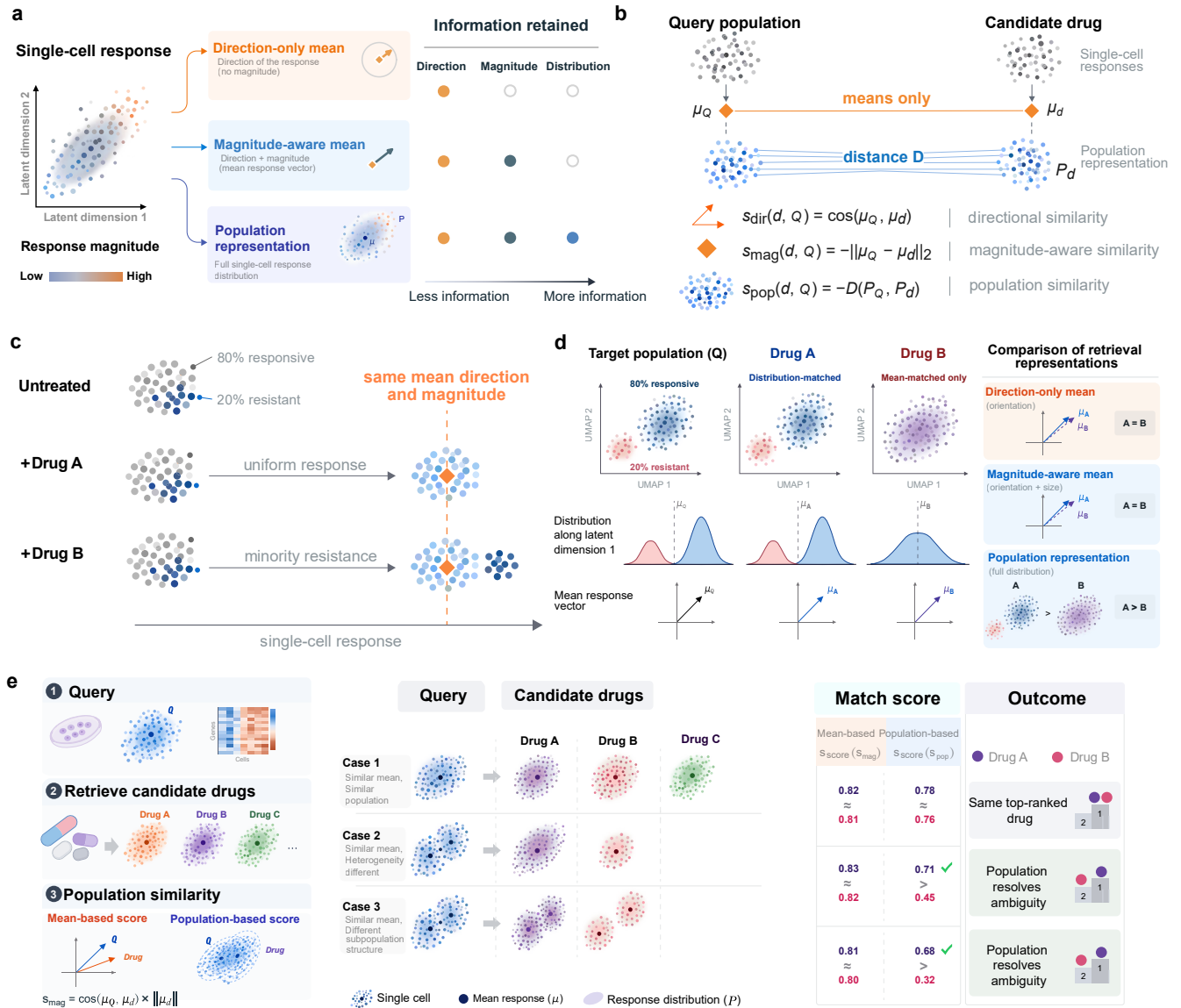


Figure 1: **Single-cell response representations retain different information and can change candidate ordering.** **a**, Three representations of the same single-cell perturbation response retain progressively more information: response direction only, direction and magnitude of the control-referenced mean response, and the full raw treated-cell response population. Filled and open circles indicate retained and discarded information. **b**, Corresponding retrieval rules compare query and candidate responses using directional similarity, magnitude-aware distance between control-referenced mean responses, or a distance between raw treated-cell response populations. **c**, Two perturbations can have the same mean response direction and magnitude while differing in their single-cell response structure. Drug A produces a relatively uniform response, whereas drug B leaves a resistant minority population. **d**, A target population and two candidates illustrate the resulting ambiguity. Drug A matches the target distribution, whereas drug B matches only its mean; the direction-only and magnitude-aware mean scores therefore tie, while the population representation prefers drug A. **e**, Population information affects retrieval only when the additional response structure changes candidate ordering. Across three illustrative cases, population-based scoring either preserves the mean-based top-ranked candidate or resolves ambiguities that are invisible to mean-based matching.

88 Mean-signature and population-level retrieval can therefore be viewed as the same ranking task  
 89 at different resolutions of the measured response. Under a shared or common reference, as within-  
 90 population variation vanishes, the control-referenced response population collapses onto its control-  
 91 referenced mean, and the energy distance between two such point masses is exactly twice the Euclidean

distance between their means, which cannot reorder candidates. Under this common-origin construction, magnitude-aware mean L2 is therefore the zero-variance limit of energy-based population retrieval (Methods); this is not a global equivalence when query and candidate matched controls differ. Direction-only cosine coarsens the mean representation further by discarding response magnitude (Fig. 1a,b). The implemented CMap-style cosine applies the same rule to the same mean vectors and is therefore identical to mean-cosine retrieval, whereas the canonical L1000 weighted connectivity score remains a distinct rank-based score (Fig. 2f). Subpopulation coverage provides an additional family of population-aware scores, interpolating from average to worst-case state matching through its aggregation parameter  $\beta$  (Methods). Together, these representations place direction-only mean matching, magnitude-aware mean matching and population comparison on an axis of increasing sensitivity to response structure (Fig. 1a,b; Supplementary Table 1).

### Most of the gain over directional matching does not require the full response population

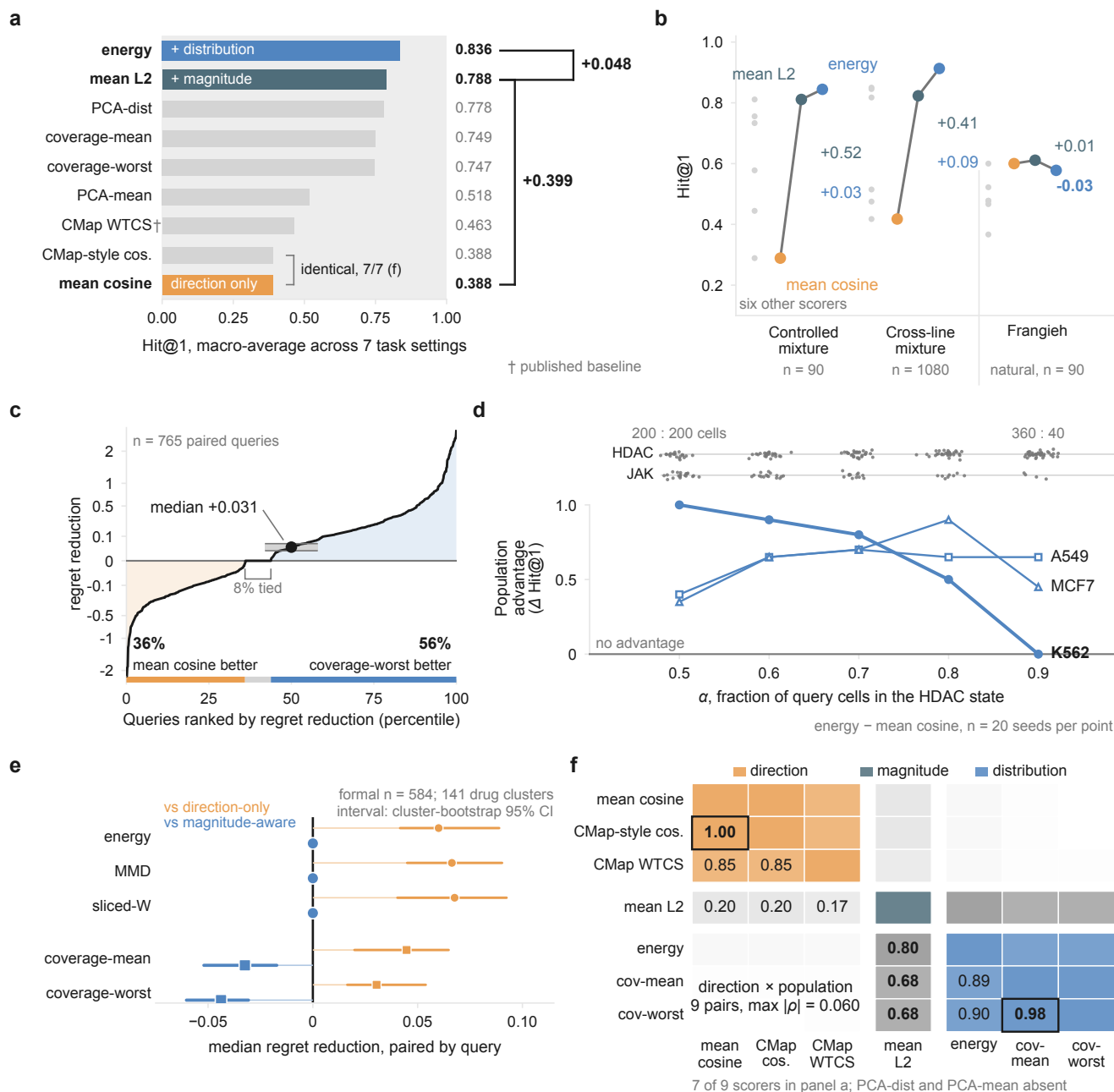
Population-level retrieval substantially outperformed direction-only mean matching, and most of that gain did not require the full response population. With candidate responses directly observed, so that the full cellular populations are available to both approaches, energy-distance retrieval reached a Hit@1 of 0.836, macro-averaged across the seven task-setting combinations analysed here, against 0.388 for mean-cosine retrieval and the numerically identical CMap-style cosine baseline<sup>11,12</sup>. Between them sits a magnitude-aware mean L2 at 0.788, applied to the same mean signatures: the step to that score is +0.399 of the gap and the step from it to the full distribution a further +0.048 (Fig. 2a). Because that first step changes both what the score reads, from direction alone to direction and magnitude, and the geometry it uses, from angular similarity to Euclidean distance, mean L2 is a magnitude-aware mean baseline rather than a pure magnitude ablation. The same ordering held in PCA space (Fig. 2a).

Frangieh provided a contrasting result. On Frangieh Perturb-CITE-seq<sup>30</sup>, the only one of the seven combinations not built from SciPlex3, the magnitude-aware mean L2 was the best of the nine scores (Hit@1 0.611), ahead of mean cosine (0.600) and of energy distance (0.578; Fig. 2b). There, retaining magnitude helped and retaining the full response distribution did not.

Population-level scoring also improved response matching under partial observation, where only part of each candidate population is available. The same rankings were judged by a regret criterion that penalizes selecting a candidate whose per-cell outcomes are worse than the best available (Methods). Across 765 queries, the worst-case subpopulation-coverage score reduced the descriptive median regret by +0.031 relative to direction-only mean cosine, with lower regret on 56% of queries. In the formal individual-drug analysis, the median reduction was +0.0304 (95% drug-cluster bootstrap CI +0.0155 to +0.0495, 720 queries from 143 drugs; Fig. 2c).

Reducing the minority response state from half of the query population to a tenth did not erode the energy-distance advantage consistently: it collapsed in one cell line and held in the other two (Fig. 2d). The pattern was shared across the population-aware scores. All five variants improved on direction-only mean cosine, whereas against the magnitude-aware mean L2 the three distributional distances have a median difference of exactly zero and both coverage scores are negative (Fig. 2e).

A reference-sensitivity analysis addressed the different biological references used by the mean and population routes. In the three cross-line tasks, where query and candidate controls can differ, energy retrieval reached Hit@1 values of 0.9189 with raw populations and 0.9158 after matched-control referencing across 2,250 queries; the corresponding MRR values were 0.9478 and 0.9462. Sliced Wasserstein changed from 0.8123 to 0.8075 Hit@1, whereas RBF-MMD and both coverage limits were unchanged



**Figure 2: A magnitude-aware mean score recovers most of the gain beyond directional scoring.** **a**, Hit@1 for nine scorers, macro-averaged over seven task-setting combinations. Direction-only mean cosine, magnitude-aware mean L2 and energy distance reach 0.388, 0.788 and 0.836, respectively. Dagger, published CMap connectivity score (WTCS). **b**, The three scorers by task ( $n = 90, 1,080$  and  $90$  queries). The magnitude step is  $+0.52$  and  $+0.41$  for the two SciPlex3 tasks and  $+0.01$  for Frangieh; the distributional step for Frangieh is  $-0.03$ . **c**, Worst-case subpopulation coverage versus direction-only mean cosine for 765 partial-observation queries. The curve shows the quantile function of per-query regret reduction, with positive values favouring coverage. Formal inference used 720 queries from 143 query-drug clusters and gave a median reduction of  $+0.0304$  (95% cluster-bootstrap CI  $+0.0155$  to  $+0.0495$ ). **d**, Energy advantage over mean cosine as a function of minority fraction  $\alpha$  ( $n = 20$  seeds per point). In K562 the advantage falls from  $+1.00$  to  $0.00$ ; in A549 and MCF7 it ranges from  $+0.40$  to  $+0.65$  and from  $+0.35$  to  $+0.45$ , respectively. **e**, Median regret reduction for five population-level scorers on 584 individual-drug queries from 141 query-drug clusters. All five scorers improve on direction-only mean cosine. Relative to magnitude-aware mean L2, the three distributional scorers have median zero and the two coverage scorers are negative ( $-0.032$  and  $-0.044$ ). **f**, Spearman correlations among seven scorers for 54,180 query-candidate comparisons from 1,260 queries. Shade,  $|\rho|$ ; rings, the closest pair within a tier. Mean L2 correlates more strongly with distributional scorers ( $\rho = 0.68$ – $0.80$ ) than with direction-only scorers ( $\rho = 0.17$ – $0.20$ ).

135 to the reported precision. The population-minus-mean effect remained positive for every scorer and  
136 cross-line setting; the Frangieh differing-control contrasts were unchanged. In partial observation, re-  
137 running query clustering and candidate assignment after control subtraction changed the rankings, but  
138 the response-matching and minority-coverage advantages remained positive (Supplementary Note 1).  
139 The controlled-mixture, GDSC2 and same-context RNA paths use a common control translation and are  
140 therefore invariant for these population distances.

141 A magnitude-aware mean score therefore recovers most of the gain beyond directional scoring, and  
142 the further increment from retaining the full population is real and much smaller. Because both criteria  
143 reward response correspondence, the next analysis tested whether the increment survives biological and  
144 functional evaluation.

### 145 **Population-level gains weaken under biological and functional evaluation**

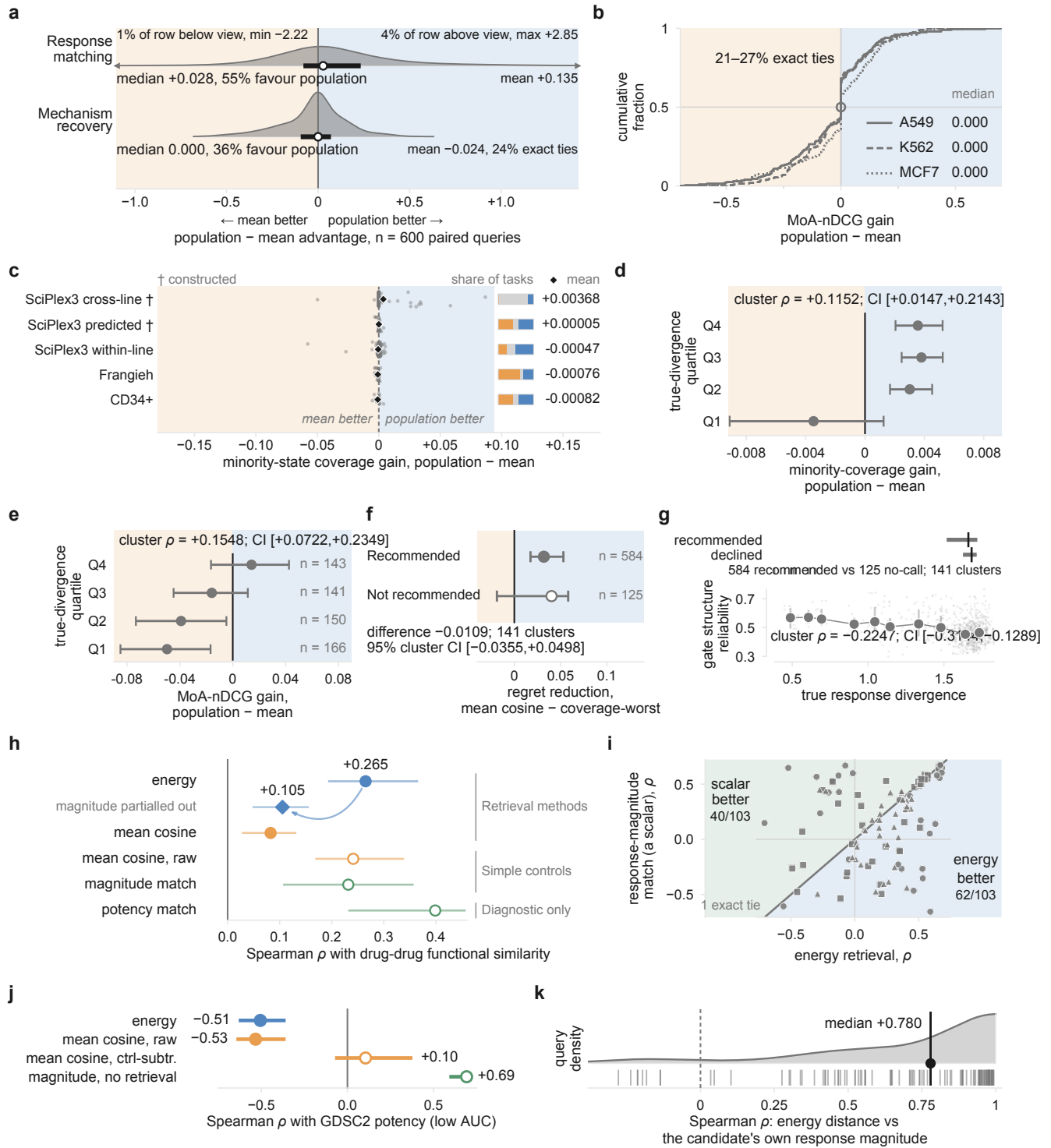
146 The value of that information depends on the evaluation criterion. We kept rankings fixed and changed  
147 only the criterion used to judge them (Fig. 3).

148 Mechanism-of-action recovery is the first task-proximal biological criterion. Among the 600 leave-  
149 drug-out queries from 130 query-drug clusters, the worst-case subpopulation-coverage scorer improved  
150 on direction-only mean cosine by a median of +0.028 under response matching (95% cluster-bootstrap  
151 CI +0.0114 to +0.0510), whereas the identical rankings judged by mechanism-of-action recovery gave  
152 a median of 0.000 (95% cluster-bootstrap CI 0.0000 to 0.0000) and a mean of -0.024 (Fig. 3a,b). At the  
153 query level, mechanism recovery decreased more often than it increased (40.8% against 35.7%). The  
154 same divergence between the two criteria appeared in all five population-level scoring variants, whose  
155 response-matching median gains span +0.028 to +0.067 while their mechanism-recovery means span  
156 -0.011 to -0.033. Gains under response matching therefore did not consistently carry over to the recovery  
157 of biologically related drugs.

158 A second task-proximal criterion tests how well population-level retrieval preserves minority-state  
159 responses. Across 239 task instances, population-level retrieval improved minority-state coverage by  
160 only +0.0011 on average, and in 102 of them the two approaches scored exactly the same (Fig. 3c).  
161 The small positive effect was concentrated in the constructed cross-cell-line tasks (+0.0037), whereas  
162 the remaining settings were near zero or slightly negative. In the formal partial-observation analysis,  
163 stratifying by measured response divergence gave coverage means from -0.0035 in Q1 to +0.0036 in Q4,  
164 with the cluster-bootstrap intervals reported in Fig. 3d and a positive cluster-bootstrap association ( $\rho$  =  
165 +0.1152, 95% CI +0.0147 to +0.2143). Under mechanism recovery, Q4 remained small and uncertain  
166 (+0.0142, 95% CI -0.0165 to +0.0425), while Q1 and Q2 were negative and Q3 crossed zero (Fig. 3e).  
167 Response divergence was therefore not sufficient to produce a more biologically useful ranking, and  
168 the pre-specified diagnostic gate did not reliably identify queries that benefit (Fig. 3f,g; Supplementary  
169 Note 1).

170 Population-level retrieval also showed an association with the external functional readout. Using  
171 GDSC2<sup>28,29</sup> dose-response profiles from cell lines excluded from the transcriptional retrieval experiment,  
172 and removing the shared cell-line sensitivity effect, energy-distance retrieval reached a Spearman  
173 correlation of +0.265 with drug-drug functional similarity, against +0.083 for the corresponding mean-  
174 signature baseline (Fig. 3h; Supplementary Table 3).

175 Response magnitude accounted for much of this association. Ranking candidates only by how closely  
176 their overall response magnitude matched the query reached +0.232 from a single scalar per perturbation,  
177 and once that magnitude channel was accounted for the remaining association for energy retrieval



**Figure 3: The population advantage weakens, and can change direction, under biological and functional evaluation.** Points show medians except quartile means in **d** and **e**. Error bars show 95% cluster-bootstrap intervals except interquartile ranges in **a**, **g** and **j**; line style in **b** and marker shape in **i** identify cell lines. **a**, 600 leave-one-drug-out queries from 130 query-drug clusters, evaluated by response matching and mechanism recovery. Median gains are +0.028 (95% CI +0.0114 to +0.0510) and 0.000 (95% CI 0.0000 to 0.0000), respectively. **b**, Mechanism-recovery gains by cell line; medians are 0.000 in all three cell lines ( $n = 200$  per cell line). **c**, Minority-state coverage gain over 239 task instances. Dots, task instances; diamonds, dataset means; daggers, constructed data. **d**, Population-minus-mean gain by response-divergence quartile. Quartiles use all 765 queries; formal inference uses 720 individual-drug queries from 143 clusters. Means are -0.0035, +0.0030, +0.0038 and +0.0036; medians are 0.0000, +0.0002, +0.0008 and +0.0015. Cluster-bootstrap Spearman correlation:  $\rho = +0.1152$ , 95% CI +0.0147 to +0.2143. **e**, The same quartiles evaluated by mechanism recovery ( $n = 600$ , 130 query-drug clusters). Means are -0.0497, -0.0391, -0.0158 and +0.0142; medians are -0.0062, 0.0000, 0.0000 and 0.0000. Cluster-bootstrap Spearman correlation:  $\rho = +0.1548$ , 95% CI +0.0722 to +0.2349. **f**, Regret reduction for 584 diagnostic-recommended and 125 no-call queries from 141 clusters. Recommended minus no-call: -0.0109 (95% cluster-bootstrap CI -0.0355 to +0.0498). **g**, Gate structure reliability versus true response divergence ( $\rho = -0.2247$ , 95% cluster-bootstrap CI -0.3144 to -0.1289); above, divergence distributions for the recommended and declined groups. **h**, Ranking against GDSC2 drug-drug similarity ( $n = 103$ ); filled, ranking methods; arc, energy partialled on magnitude. **i**, Per-query comparison of energy distance with the magnitude scalar; energy wins on 62 queries and loses on 40. **j**, Rankings against absolute GDSC2 potency. Energy and raw mean cosine are negative (-0.506 and -0.533); control-subtracted mean cosine and magnitude alone are positive (+0.105 and +0.692). **k**, Per-query Spearman correlation between energy distance and candidate magnitude; median +0.780.

was +0.105, positive in all three cell lines (Fig. 3h,i). Scored against absolute GDSC2 potency instead, energy and raw mean cosine turn negative, since matching a weak query's magnitude returns other weak drugs, whereas ranking by candidate magnitude alone turns positive (Fig. 3j). The channel is also directly measurable: across queries the query–candidate energy distance tracks the candidate's own response magnitude at a median Spearman correlation of +0.780 (Fig. 3k). Population-level retrieval therefore carries external functional signal, most of which is recoverable from response magnitude.

The value of population-level retrieval therefore depends on how success is measured, including criteria beyond transcriptomic response matching.

### **Evaluator form can reverse the preferred retrieval strategy, and natural heterogeneity leaves little reordering headroom**

We tested this dependence with a biological measurement unavailable to the retrieval methods.

Frangieh Perturb-CITE-seq<sup>30</sup> measures RNA and 20 surface proteins in the same 218,331 cells, so an RNA-derived ranking can be judged by a modality withheld from every retrieval method. We built two protein-based evaluators from those same measurements, changing only how they were summarized, and scored the identical two RNA-derived rankings against both (Fig. 4a). The first compared the mean protein responses of each query–candidate pair, and under it mean-signature retrieval outperformed energy-distance retrieval (+0.242 versus +0.222).

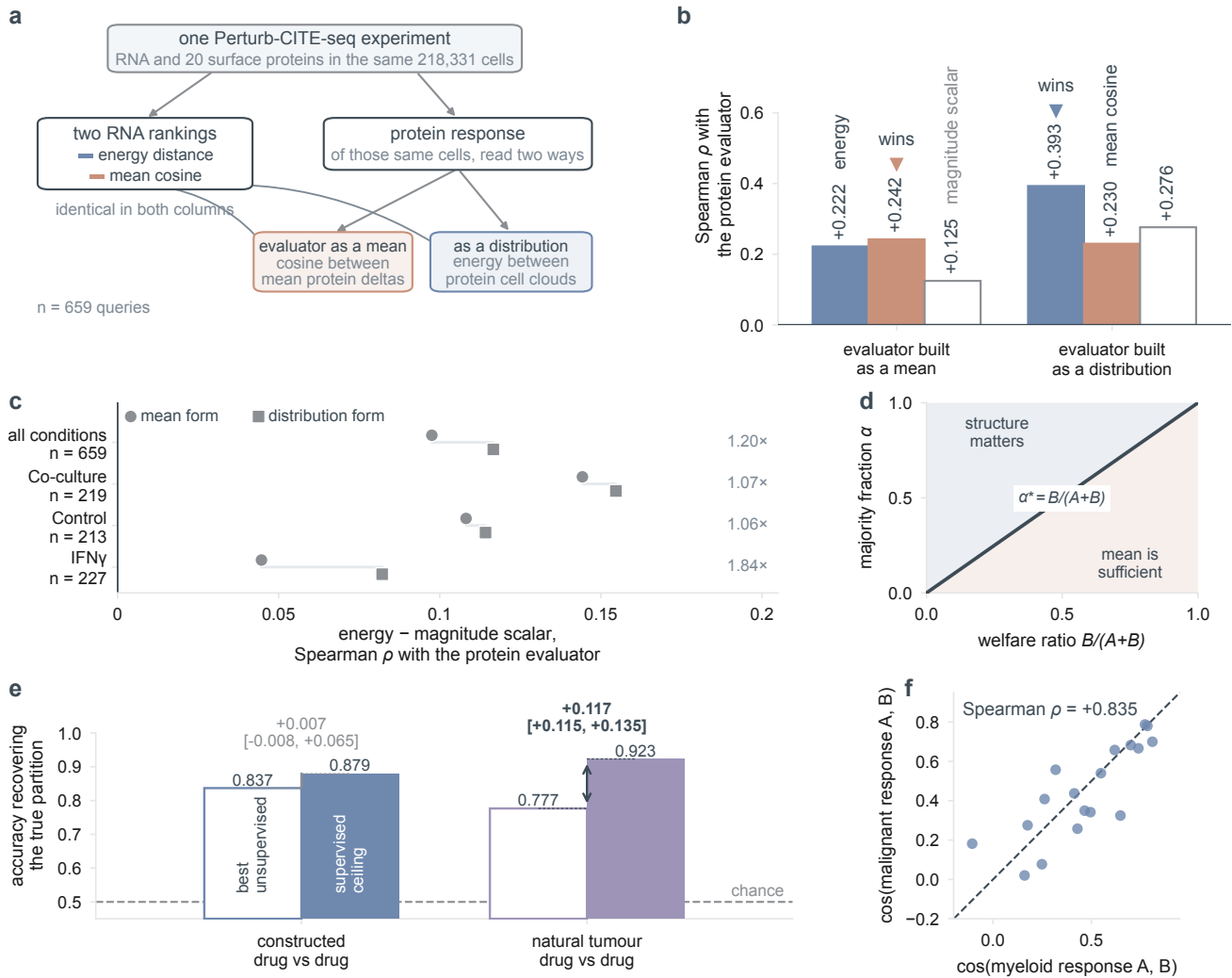
Against the full distributions of the same protein responses the ordering reversed: energy-distance retrieval reached +0.393 against +0.230 for mean-signature retrieval (Fig. 4b). Energy led under the distribution-based evaluator in all three immune conditions, and the change in which method wins occurred in two of the three. Changing only how the same held-out biological measurements were represented was therefore enough to reverse which retrieval strategy performed better.

Part of that reversal needs no distributional comparison. A scalar score that compares only the overall magnitude of the query and candidate responses rose from +0.125 under the mean-based protein evaluator to +0.276 under the distribution-based one. Energy distance stayed ahead of that control under both forms, by +0.098 and +0.117 and in every immune condition separately (Fig. 4c). Response magnitude therefore explains part of the reversal, while energy retains an additional association beyond that scalar channel. The same held-out measurements can favour different retrieval strategies depending on how their response structure is represented.

A controlled HIR-Bench construction showed that state-specific preferences can change the preferred candidate; its analytical boundary did not transfer to the biological datasets (Fig. 4d; Supplementary Note 2).

We tested whether state-dependent perturbation responses are recoverable in naturally heterogeneous tissue<sup>16</sup>. In glioblastoma samples from ZhaoSims2021, the malignant and myeloid compartments of each tumour were analysed separately against compartment-matched controls. A supervised probe recovered within-patient drug-response identity at an accuracy of 0.923 against 0.777 for the best unsupervised procedure, a median paired gap of +0.117 against +0.007 in a constructed comparator (Fig. 4e; Supplementary Note 3).

Whether that structure is large enough to change which drug should be preferred is a separate question, and the tumour data constrain it. Across 18 within-patient drug pairs, myeloid-response similarity ranked malignant-response similarity at Spearman  $\rho = +0.835$  across the two disjoint compartments, which share no cells (Fig. 4f). The coefficient describes this cohort. Drug-response similarity was largely preserved across distinct cellular compartments, leaving less room for state-specific structure to reorder



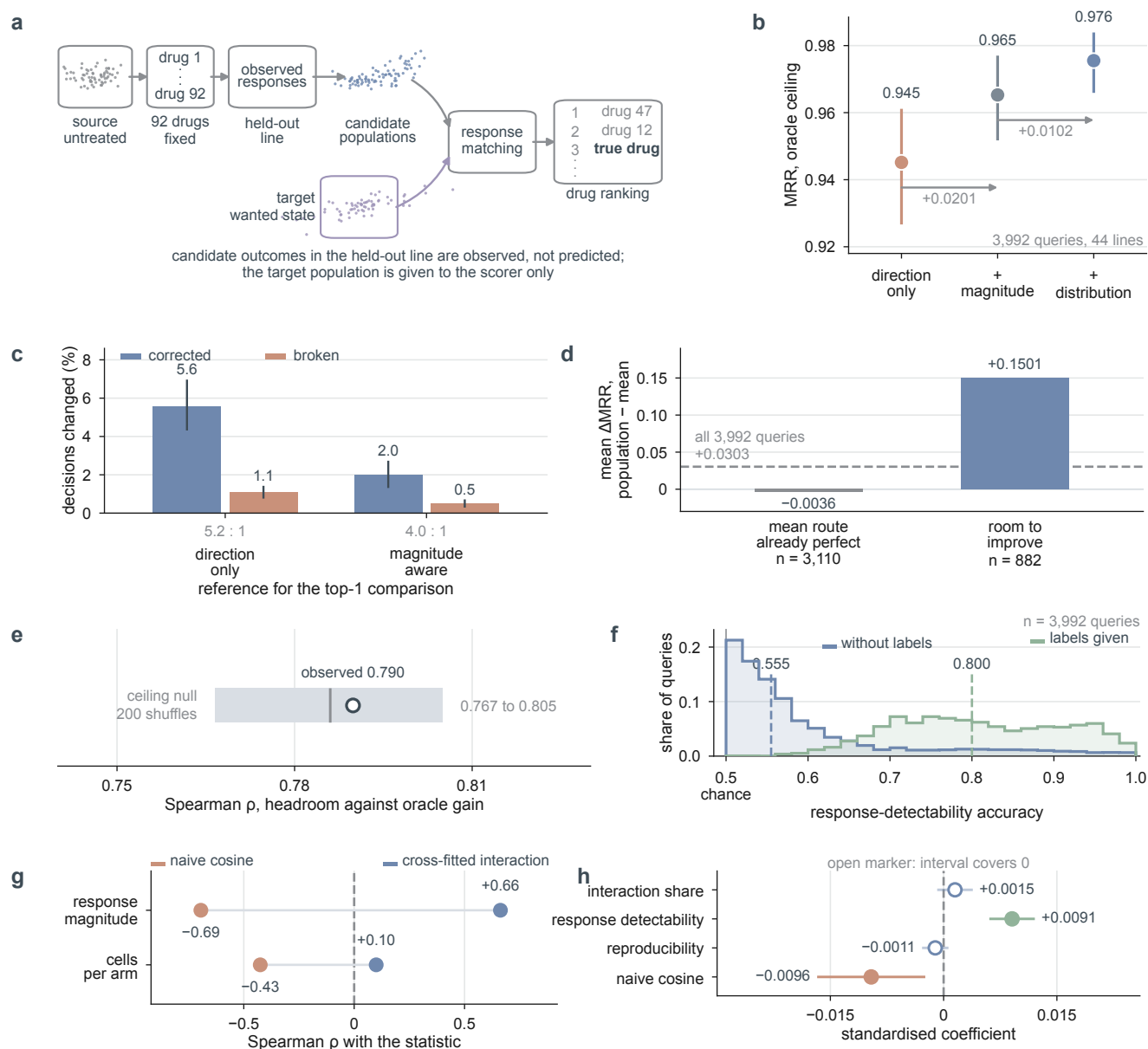
**Figure 4: External evaluator form changes the preferred retrieval strategy, whereas heterogeneity need not reorder candidates.** **a–c**, The same RNA-derived rankings evaluated using mean-based or population-based protein responses. **a**, Perturb-CITE-seq measurements of RNA and 20 surface proteins in 218,331 cells; protein responses are centred on matched controls. **b**, Spearman  $\rho$  of three RNA-side scores against the mean-based and population-based protein evaluators. Each bar is the median over  $n = 659$  queries of the per-query rank correlation across candidate knockouts. Open bar, response-magnitude control. **c**, Energy relative to the magnitude control under each evaluator, pooled and by immune condition; ratios range from 1.06 to 1.84. **d**, HIR-Bench boundary  $\alpha^* = B/(A+B)$  separating settings in which the mean is sufficient from those in which subpopulation structure can change the preferred candidate. **e, f**, Four glioblastoma tumours analysed in disjoint malignant and myeloid compartments with matched drugs and controls. **e**, Recovery of within-patient drug-response identity; open bars, best unsupervised method; filled bars, supervised probe. Median paired gap, +0.117 (95% patient-level cluster-bootstrap CI +0.115 to +0.135,  $n = 36$  splits), versus +0.007 in the constructed comparator. **f**, Myeloid-response similarity versus malignant-response similarity across compartments (Spearman  $\rho = +0.835$ ,  $n = 18$  drug pairs); dashed line, equal similarity.

221 candidates than the presence of heterogeneity alone would suggest. This measures preservation of drug  
 222 ordering; the intervention-identification task below tests whether cellular structure changes candidate  
 223 preference.

#### 224 Population information changes an intervention decision when candidate responses are observed

225 A retrieval system that proposes interventions must rank compounds whose effect in the query context  
 226 has not been measured. We evaluated this problem as a source-to-target intervention-identification task  
 227 on Tahoe-100M<sup>15</sup>: a held-out cell line’s untreated cells and a desired treated population enter, and a  
 228 ranking of a fixed 92-compound library leaves, with the true intervention’s rank as the outcome (3,992  
 229 queries over 44 cell lines; Fig. 5a). The task measures intervention identification from an observed re-  
 230 sponse produced by one library compound. It was run twice, once with candidate responses observed

231 within the query line and once with them predicted from the other 43 lines, separating the value of  
 232 population information from the cost of forecasting it.



**Figure 5: Population structure adds a small oracle retrieval gain beyond response magnitude.** Source-to-target intervention identification on Tahoe-100M: 3,992 queries over 44 cell lines and a fixed 92-compound library, with candidate responses observed within the query line; predicted candidates are evaluated in Fig. 6. **a**, A held-out line's untreated cells and a desired treated population define the query; candidates comprise all 92 compounds. **b**, Mean reciprocal rank for the three scoring rules in Fig. 1a, with 95% bootstrap intervals: 0.945, 0.965 and 0.976. The two increments are +0.0201 and +0.0102. **c**, Top-1 decisions changed relative to each mean reference; ratios show the relative frequency of the two directions. **d**, The pooled +0.0303 advantage stratified by whether the mean route leaves headroom; on 3,110 of 3,992 queries it already ranks the true drug first at every seed. **e**, Spearman correlation between mean-route headroom and population gain, with a within-context permutation null. **f**, Single-cell response detectability for a query drug versus vehicle cells from the same line, using up to 300 cells per arm; median accuracy 0.800 with labels and 0.555 without labels. **g**, Interaction statistics versus response magnitude and cells per arm. The naive  $1 - \cos(r_1, r_2)$  gives -0.69 and -0.43; cross-fitted  $S_{\text{int}}$  gives +0.66 and +0.10. **h**, Standardised coefficients from regressions of population-route reciprocal rank, with mean-route reciprocal rank as a covariate; the first three rows use  $n = 3,618$  queries in 44 cell-line clusters and the naive cosine uses  $n = 3,620$ . Filled markers, intervals excluding zero.

233 With candidate responses observed, the three scoring rules of Fig. 1a again form a ladder. Mean re-  
 234 cipocal rank was 0.945 for direction-only mean cosine, 0.965 once response magnitude was restored and  
 235 0.976 for the energy distance between populations, so the step to the magnitude-aware mean is +0.0201

and the step from it to the population  $+0.0102$  (Fig. 5b). That last step is not a property of population scoring as such. Of the two secondary population scores specified for this task, maximum mean discrepancy reproduces the energy result (MRR 0.975,  $+0.0101$  over the magnitude-aware mean), whereas sliced Wasserstein distance reaches 0.922 and sits below both mean baselines ( $-0.0230$  against direction-only matching, 95% CI  $-0.0301$  to  $-0.0166$ ;  $-0.0431$  against the magnitude-aware mean,  $-0.0509$  to  $-0.0359$ ), so two of the three population scores clear both mean baselines here and one does not. The advantage reaches individual decisions: against the magnitude-aware mean, population scoring put the true drug first on 2.0% of queries where the mean route did not and broke a correct choice on 0.5%, a fourfold asymmetry, and against direction-only matching the two rates are 5.6% and 1.1% (Fig. 5c).

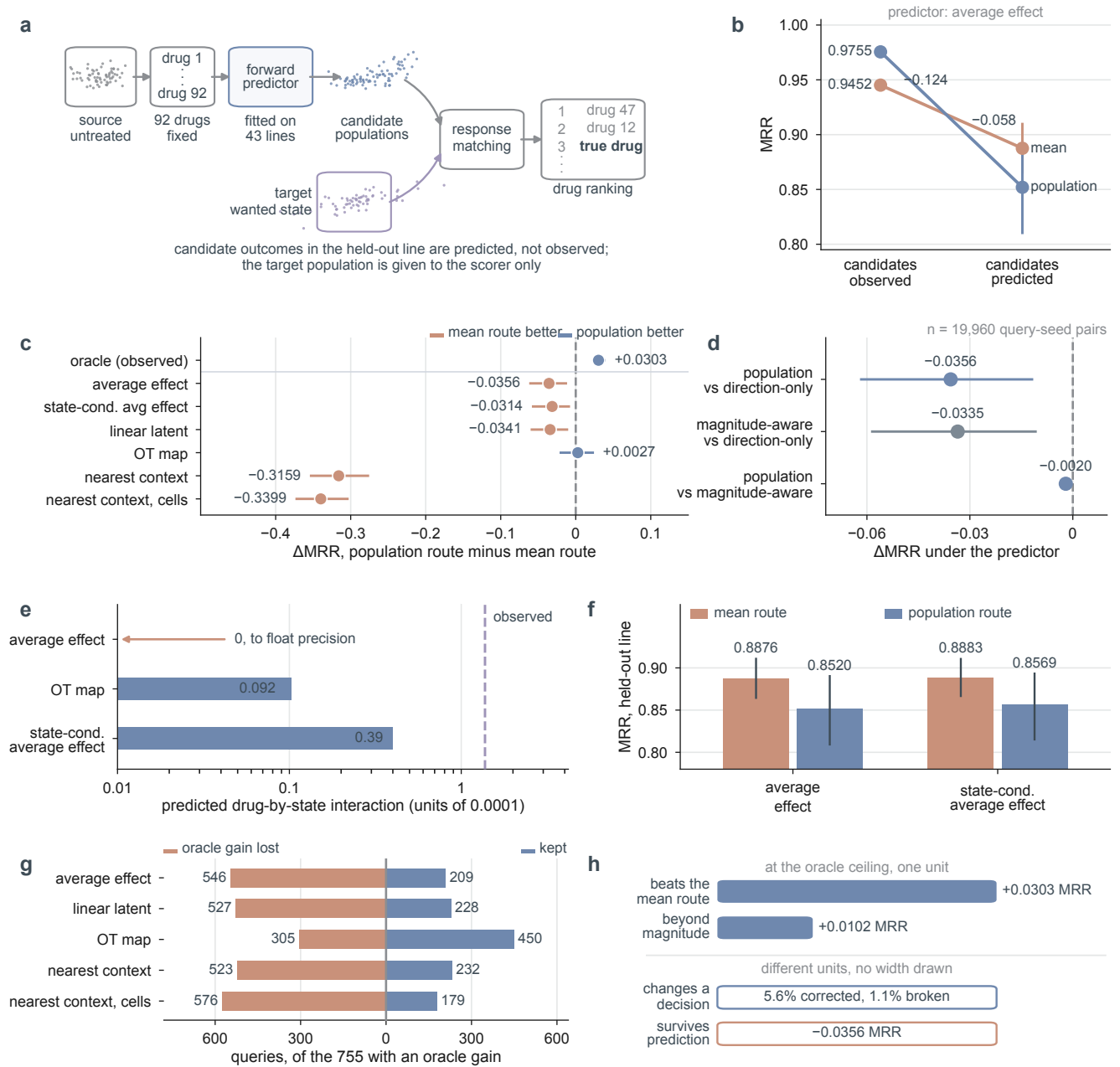
The pooled oracle gain is limited by the headroom that mean matching leaves. On 3,110 of the 3,992 queries the direction-only mean route already ranks the true drug first at every seed, where the population route can only lose and does, by  $-0.0036$  on average; the remaining 882 queries more than account for the pooled  $+0.0303$ , gaining  $+0.1501$  (Fig. 5d). Headroom of that kind correlates with the population route’s gain at Spearman  $\rho = +0.79$ , but a permutation null that preserves the ranking ceiling reproduces it at  $+0.786$  ( $+0.767$  to  $+0.805$ ; Fig. 5e, Methods), so the correlation measures the room available for improvement rather than which queries benefit.

We next tested which measurable query properties explain the remaining advantage. A query drug’s own treated cells are separable from vehicle cells of the same line at a median accuracy of 0.800 when the labels are given and 0.555 without them, on a balanced two-class problem where 0.5 is chance (Fig. 5f), and drug-by-state interaction was measured by a cross-fitted statistic that is zero in expectation under an additive null and largely free of the cell-count and magnitude dependence a naive response cosine carries (Fig. 5g; Methods). Conditional on the mean route’s own reciprocal rank, response detectability is the only one of the three whose interval excludes zero ( $+0.0091$ , 95% CI  $+0.0060$  to  $+0.0121$ ,  $n = 3,618$  queries in 44 cell-line clusters); neither the interaction share ( $+0.0015$ ,  $p = 0.21$ ) nor its reproducibility across disjoint halves ( $-0.0011$ ,  $p = 0.21$ ) is associated with the population route. Substituting the naive cosine for the cross-fitted statistic reverses the sign of the association ( $-0.0096$ , 95% CI  $-0.0167$  to  $-0.0024$ ; Fig. 5h). Population-level information therefore has measurable value at the oracle, but limited room in which to act.

## **The oracle advantage does not survive forward prediction**

Replacing the observed candidate responses with predicted ones removes the advantage and reverses its sign (Fig. 6a). The primary average-effect predictor costs the mean route 0.058 MRR and the population route 0.124, and the two cross (Fig. 6b). Under it the population route sits  $-0.0356$  below direction-only mean matching (95% CI  $-0.0620$  to  $-0.0115$ ), so the oracle gain does not attenuate but changes sign (Fig. 6c). None of the six forward predictors evaluated delivers a population advantage, and the Gaussian optimal-transport map is the only one whose point estimate is not negative ( $+0.0027$ , interval covering zero).

The dominant deficit is a magnitude error rather than a distribution-specific one. On the same predictions the magnitude-aware mean loses  $-0.0335$  to direction-only matching, almost the whole of the population route’s deficit, although it reads no distribution at all; against the magnitude-aware mean itself the population route sits only  $-0.0020$  below ( $-0.0036$  to  $-0.0005$ ; Fig. 6d). An additive predictor translates the query line’s own vehicle cells by an effect averaged over the other 43 lines, so it reproduces that population’s heterogeneity exactly while generating no drug-by-state interaction at all, an algebraic property of the model class rather than a training failure (Fig. 6e; Supplementary Note 4).



**Figure 6: Forward prediction removes the oracle population-retrieval advantage.** The task, queries, library and split are as in Fig. 5; candidate responses in the held-out line are produced by a forward predictor fitted on the other 43 lines. **a**, The intervention-identification task with predicted candidate responses. **b**, Both routes travelling from observed to predicted candidates under the primary average-effect predictor; the mean route gives up 0.058 MRR and the population route 0.124, and the two cross. **c**,  $\Delta$ MRR of the population route over direction-only mean matching, at the oracle and for six predictors; under an additive predictor the oracle gain inverts rather than attenuating. **d**, The same predictions judged three ways: the population route's  $-0.0356$  deficit against direction-only is matched by the magnitude-aware mean's  $-0.0335$ , and  $-0.0020$  remains once magnitude is controlled. **e**, Median drug-by-state interaction from each predictor versus the cross-fitted value measured in real cells (dashed); the additive predictor generates zero interaction. **f**, A state-conditioned average-effect predictor generates interaction while preserving mean-route accuracy, but the population route remains lower. **g**, Among the 755 queries on which the oracle population route had any gain, 209 retain a gain under the primary average-effect predictor and 546 do not. Additional gains occur among queries with no oracle gain: 343 for the primary predictor and 1,165 for the Gaussian OT map. **h**, MRR budget across Figs. 5 and 6; the first two rows share the oracle MRR scale, whereas the final two show a rate and a signed gain.

279 A state-conditioned average-effect predictor is the first to generate real interaction while preserving  
280 the mean route's MRR (0.8883 against 0.8876), and it still leaves the population route behind ( $-0.0314$ ,  
281  $-0.0584$  to  $-0.0076$ ; Fig. 6f). Generating state-specific interaction is therefore necessary for the popula-  
282 tion route but not sufficient. The same attrition is visible query by query: of the 755 queries on which  
283 the oracle population route had any gain, the average-effect predictor keeps one on 209 and loses it on  
284 546 (Fig. 6g).

285 Together the two arms locate where population-level information enters and where it is lost (Fig. 6h).  
286 With responses observed it is present, reaches the ranking decision, and is bounded by the performance  
287 of mean signatures. With responses predicted it does not survive: the predictors that best preserved  
288 mean-level retrieval were additive and generated no state-specific perturbation response, whereas state  
289 conditioning restored that interaction without restoring a population-level advantage.

## 290 Discussion

291 The results trace a chain of attrition. Response populations carry retrievable information; a smaller part  
292 exceeds response magnitude; a smaller part changes which candidate is preferred; and almost none  
293 survives once candidate responses are predicted rather than observed. The binding constraint changes  
294 along that chain: with observed candidates it is the ranking headroom left by mean matching, and with  
295 unobserved candidates it is the fidelity of the forward predictor.

296 Single-cell perturbation atlases<sup>13–15</sup> change the representation available for drug retrieval: a pertur-  
297 bation can now be described by a population of cellular responses rather than by a single averaged  
298 signature. Across the datasets and evaluators examined here, retaining the full population increased  
299 the information available to the retriever, but only a smaller component beyond response magnitude  
300 reached the ranking, and it did so under response matching, where Fig. 2e shows that it adds nothing to  
301 the magnitude-aware mean under the regret criterion; under mechanism-of-action recovery the whole  
302 population advantage does not survive (Fig. 3a,b,e). The central distinction is therefore not mean versus  
303 distributional retrieval, but information that is available versus information that changes the decision.  
304 Classical perturbation signatures and single-cell populations are points on a continuum of retained re-  
305 sponse structure, and the relevant question is how much of that structure the decision at hand requires.

306 This distinction has direct implications for artificial intelligence virtual cells<sup>1,2</sup>. Accurate prediction of  
307 the average perturbation response is not sufficient for population-level intervention selection. A model  
308 that applies the same perturbation effect to every cell preserves heterogeneity in the starting population  
309 but generates no state-specific differential response; this limit is exact for strictly additive predictors,  
310 whereas more flexible latent models are not algebraically constrained to it (Supplementary Note 4). A  
311 population-aware retriever cannot recover variation absent from the predictor. State-dependent differ-  
312 ential response is required to exploit that source of population information, but it is not sufficient: state  
313 conditioning restored measurable drug-by-state interaction while preserving mean-level performance,  
314 yet did not restore a population-level retrieval advantage (Fig. 6e,f). The magnitude-aware mean loses  
315 almost exactly what the population route loses, identifying predicted response magnitude as the domi-  
316 nant remaining error rather than population shape (Fig. 6d).

317 The response and interaction diagnostics did not identify a reliable query-level rule for when popu-  
318 lation scoring helps, and state-dependent differential response represents one of several possible forms  
319 of population structure. The statistical representation of an external readout can change which method  
320 ranks higher<sup>27,31</sup>. Evaluation should therefore test whether added cellular information changes an out-  
321 come relevant to the intended use and compare it with simpler channels such as response magnitude<sup>23</sup>.

322 The functional readouts analysed here measure external association rather than therapeutic benefit. A  
323 richer cellular measurement supports a richer retrieval method when the information it adds survives  
324 to the decision that method is intended to make.

## 325 **Methods**

### 326 **Study design and data resources**

327 This study tests when retaining the full population of single-cell perturbation responses improves re-  
328 trieval beyond a conventional mean expression signature. The analyses cover four roles: observed-  
329 population retrieval, evaluation against biological and functional criteria, measurement of naturally  
330 occurring response heterogeneity, and assessment of when population-level information can affect in-  
331 tervention selection. These roles address distinct questions and are reported separately.

332 The primary observed-population retrieval analyses used SciPlex3<sup>14</sup>, comprising 276,325 cells across  
333 three cancer cell lines (A549, K562 and MCF7), 188 compounds shared across the three contexts and  
334 a 2,000-gene expression space. SciPlex3 supports both within-context retrieval and controlled cross-  
335 context constructions because the same compound library is measured across all three cell lines. Frang-  
336 ieh melanoma Perturb-CITE-seq<sup>30</sup> spans Control, IFN $\gamma$  and Co-culture conditions with 239 CRISPR  
337 knockouts. Both analyses read the same RNA matrix under two filters: the retrieval analysis uses the  
338 218,023 cells left after dropping nine knockout targets with fewer than 50 cells, and the evaluator-shape  
339 experiment uses all 218,331 cells of the RNA–protein barcode intersection, of which 215,915 contribute to  
340 at least one condition mean once the 60-cell floor per knockout is applied. Frangieh provides an indepen-  
341 dent perturbation-retrieval setting and matched surface-protein measurements for the evaluator-shape  
342 experiment, in which the protein modality is withheld from every retrieval method and used for evalu-  
343 ation. This is a genetic-perturbation test of the retrieval formulation. CD34+ haematopoietic progenitor  
344 data (GSE306429)<sup>32</sup> contain 33,984 cells and 36 compounds and are used as a negative-control setting for  
345 population-level retrieval.

346 Three additional resources serve roles outside the primary retrieval benchmark. ZhaoSims2021<sup>16</sup>  
347 contains drug-treated patient-derived glioblastoma tissue and supplies natural-tissue diagnostics of  
348 state-dependent responses and their recoverability. Tahoe-100M<sup>15</sup> provides larger-scale measurements  
349 of the same information requirements under intrinsic within-cell-line heterogeneity and is the substrate  
350 for the source-to-target intervention-identification benchmark of Figs. 5 and 6. We analyse plate 3 only,  
351 which comprises 4,158,278 cells across 50 cell lines and 93 compounds, using the released 2,304-gene  
352 log1p-transformed representation without refitting the preprocessing pipeline. Finally, GDSC2 dose-  
353 response profiles provide an external functional readout for drug–drug similarity and are joined to the  
354 SciPlex3 compound panel independently of the transcriptomic retrieval scores.

355 Observed responses, predicted responses, external functional measurements and natural-tissue anal-  
356 yses address distinct questions in this study. Dataset scale, inclusion criteria, experimental role and  
357 query construction are summarized in Supplementary Table 2.

### 358 **Preprocessing and construction of perturbation-response populations**

359 Single-cell expression analyses were performed in log-transformed expression space using the processed  
360 gene sets defined for each dataset. SciPlex3, Frangieh and CD34+ were analysed in 2,000-gene represen-  
361 tations. For SciPlex3 and Frangieh these genes were selected in this study from raw counts: counts were  
362 normalized to a total of  $10^4$  per cell and log1p-transformed, 2,000 highly variable genes were chosen by  
363 the Seurat criterion<sup>33</sup> on a subsample of cells, and the final matrices were then rebuilt from raw counts

on those genes using all-gene size factors so that normalization is applied once. SciPlex3 was additionally downsampled to at most 120 cells per compound, cell line and dose, and 3,000 control cells per cell line. The CD34+ tensor was received in processed form and the feature-selection procedure applied to it is not documented by the source. For ZhaoSims2021, counts were normalized to a total of  $10^4$  per cell, log1p-transformed and reduced to 2,000 highly variable genes. Tahoe-100M was used in the released preprocessed form (2,304 highly variable genes, log1p transformed), with no additional feature selection or representation fitting. Dataset-specific filtering and minimum-cell requirements for individual analyses are given in the corresponding Methods sections below.

For a perturbation  $d$  measured in cellular context  $c$ , let

$$X_{d,c} = \{x_1^{(d,c)}, \dots, x_{n_{d,c}}^{(d,c)}\}$$

denote its treated single-cell expression profiles and let

$$\mu_{0,c} = \frac{1}{n_{0,c}} \sum_{j=1}^{n_{0,c}} x_j^{(0,c)}$$

denote the mean of the matched control population. The conventional perturbation signature is the control-referenced mean response

$$\bar{d}_{d,c} = \frac{1}{n_{d,c}} \sum_{i=1}^{n_{d,c}} x_i^{(d,c)} - \mu_{0,c}.$$

Mean-signature retrieval operates only on  $\bar{d}_{d,c}$ .

Mean-signature retrieval uses the control-referenced mean response, whereas population-level retrieval scores receive the raw normalized-expression treated-cell matrices,

$$P_{d,c} = \left\{ x_i^{(d,c)} \right\}_{i=1}^{n_{d,c}},$$

while variation among cells remains available to the population score. Mean cosine and mean  $L_2$  each subtract the corresponding control mean from the treated-cell mean; query and candidate controls may therefore differ. Energy distance, RBF-MMD, sliced Wasserstein and coverage use the raw normalized-expression populations and do not subtract matched controls in the manuscript population-scoring path. A control-referenced population comparison would constitute a separate sensitivity construction and is not part of the reported population-scoring path.

Controls were matched to the biological context in which the perturbation was measured. Analyses comparing cellular states or compartments reference each state to its own matched control population. This state-specific construction is used for the differential-response analyses in predicted populations, patient-derived glioblastoma and Tahoe-100M and is defined in the corresponding sections below.

Cell sampling followed definition of the response populations. Minimum cell numbers, per-condition caps and random seeds are reported with the corresponding analysis. Protein measurements used for the evaluator-shape experiment were processed separately, including barcode-based RNA-protein matching and protein-specific normalization, as described below.

### Population-to-population retrieval and signature baselines

For a query response population  $Q$  and a library of candidate response populations  $\{P_d\}_{d=1}^D$ , retrieval assigns each candidate a similarity or distance to the query and orders the candidate library accordingly.

Population-level methods operate directly on the empirical raw normalized-expression populations defined above, whereas signature-based methods first reduce each population to its control-referenced mean response. Query construction and candidate libraries are kept identical when comparing the two representations.

**Population-level retrieval.** We evaluated energy distance<sup>34</sup> (the primary population score), maximum mean discrepancy<sup>35</sup>, sliced Wasserstein distance<sup>36</sup>, and the mean and worst-case limits of a subpopulation-coverage score. All rank candidates from smallest to largest distance. For a candidate population  $P = \{x_1, \dots, x_m\}$  and a query population  $Q = \{y_1, \dots, y_n\}$  the energy distance is estimated as

$$\hat{\mathcal{E}}(P, Q) = \frac{2}{mn} \sum_{i,j} \|x_i - y_j\| - \frac{1}{m(m-1)} \sum_{i \neq i'} \|x_i - x_{i'}\| - \frac{1}{n(n-1)} \sum_{j \neq j'} \|y_j - y_{j'}\|,$$

using the Euclidean norm. Excluding the self-pairs makes this the unbiased U-statistic form; the biased V-statistic divides the two within-population terms by  $m^2$  and  $n^2$  instead, which adds each population's own mean within-population distance divided by its cell count and therefore shifts candidates of different size or dispersion by different amounts. All reported energy distances use the U form except the PCA-latent population score of Fig. 2a, which is computed by a separate routine in the V form; the two forms are compared arm by arm in the code release. Energy distance and RBF-MMD use the unbiased U-statistic by default; sliced Wasserstein has no U/V choice. Coverage uses the same U default for its energy/MMD component distances. The PCA-latent population diagnostic is the explicitly labelled legacy V-statistic exception. Population distances are computed on raw normalized expression; mean-signature scores subtract the matched control mean. Fixed cell caps, penalties for undersized populations and the implementation settings for secondary scores are reported in Supplementary Methods.

**Subpopulation-coverage score.** The query is partitioned into  $K$  components ( $k = 2$  throughout). In construction-labelled experiments, the known construction labels are used directly when they are carried by the task. In the controlled-mixture ranker, candidate labels are unavailable and candidate cells are assigned to the nearest target-component mean by Euclidean distance. In partial-observation experiments, where no construction labels are available, the query partition is  $k$ -means on the query cells with three initializations and a fixed seed, and each candidate cell is assigned to the query centroid with the highest cosine similarity after row normalization. Writing  $d_k$  for the base distance between the query cells and the candidate cells of component  $k$ , the coverage distance and its ranking score are

$$D_\beta = \frac{1}{\beta} \left( \log \sum_{k=1}^K e^{\beta d_k} - \log K \right), \quad S_\beta = -D_\beta.$$

The base distance  $d_k$  is one of energy, MMD or sliced Wasserstein, so coverage is a family built on the same kernels rather than an independent metric. A candidate that populates no cell of a query component receives the penalty distance for that component, which is how the score charges a candidate for leaving part of the query uncovered.

As  $\beta \rightarrow 0$ ,  $D_\beta$  averages subpopulation distances; as  $\beta \rightarrow \infty$ , it approaches the worst-matched component. For  $K = 1$ , it equals the global distance. Coverage-mean and coverage-worst are these two limits; numerical implementation details are given in Supplementary Methods.

**Mean-signature baselines.** For each response population  $P_{d,c}$ , the corresponding mean signature is the control-referenced mean response  $\bar{d}_{d,c} = \text{mean}(P_{d,c}) - \mu_{0,c}$ . Mean-signature retrieval ranks candidates using either cosine similarity or Euclidean distance between these vectors. The implemented CMap-style cosine baseline applies cosine similarity to the same control-referenced mean differential-expression vectors and is therefore numerically identical to mean-cosine retrieval (Spearman  $\rho = 1.000$  over 54,180 query–candidate scores). The canonical L1000 weighted connectivity score (WTCS)<sup>12</sup> is included as a separate signature-based baseline. WTCS uses rank-based enrichment of the mean-collapsed expression response and is related to, but not algebraically identical to, the cosine implementation used here; it gives a different Hit@1 (0.4635).

To test whether the comparison depended on the original gene space, we also evaluated a PCA representation. One basis was fitted per retrieval query to a uniform sample of up to 4,000 cells pooled from the query and all 43 candidate populations, including the correct candidate, and reduced to 30 components. The fit was transductive and unsupervised: it used expression values but no candidate identity, ground-truth label or subpopulation label. Query cells were disjoint from the correct candidate’s cells. Control cells did not enter the fit; each control mean was projected through the fitted basis. The same comparison was then performed either between latent-space means or between the corresponding latent response populations.

**Connection between mean and population retrieval.** Under a shared or common reference, define the control-referenced response population  $R_{d,c} = P_{d,c} - \mu_{0,c}$ , and write it as

$$R = \mu + \lambda\epsilon, \quad E[\epsilon] = 0,$$

where  $\mu = \bar{d}_{d,c}$  is the control-referenced mean response,  $\epsilon$  represents within-population variation and  $\lambda$  controls its magnitude. As  $\lambda \rightarrow 0$ , the response population collapses to a point mass at  $\mu$ , and population comparison therefore loses all information beyond the mean. The mean *representation* is the endpoint of this variance-sensitivity construction. For two point masses, the energy distance is exactly  $2\|\mu_Q - \mu_d\|_2$ , so its ranking is the same as the magnitude-aware mean  $L_2$  ranking up to a positive constant, not the cosine ranking. In this common-origin construction, subtracting  $\mu_{0,c}$  from both populations is a common translation and does not change the population distance. In the reported population-scoring path, query and candidate controls can differ, so the corresponding translations are not common and no global algebraic equivalence with mean  $L_2$  follows. The cosine is a further coarsening of the mean representation that discards  $\|\mu_Q - \mu_d\|_2$  altogether. Figure 2a measures what that coarsening costs.

The continuum refers to the amount of within-population response structure available to the scorer, not to an equivalence between otherwise different scoring functions: WTCS remains a distinct rank-based signature score, and energy, MMD and sliced Wasserstein retain their own distance definitions.

## Response-matching and partial-observation evaluation

**Observed-population retrieval.** When complete candidate response populations were available, retrieval was evaluated over heterogeneous queries. Each query is a population built by combining two response modes at mixing weight  $\alpha \in \{0.5, 0.7, 0.9\}$ , and the correct answer is the candidate that reproduces both modes at the same weight, built from the held-out half of the same cells so that no cell is shared between a query and its answer. Two further candidates carry one mode each, and 40 distractor populations complete a 43-candidate library.

The three tasks differ in what the two modes are. In the controlled SciPlex3 task they are the HDAC-

and JAK-associated response sources within a single cell line at 10  $\mu$ M, each pooling every compound annotated to that class, so the two single-mode candidates are HDAC-only and JAK-only populations and each distractor is an individual compound of another class. In the cross-line and Frangieh tasks they are one perturbation observed in two biological contexts, two cell lines or Control and IFN $\gamma$ , so the two single-mode candidates are that perturbation observed in one context only and each distractor is a different perturbation mixed across the same two contexts at the same weight; cross-line perturbations pool all measured doses. The two single-mode candidates are each referenced to their own single-context control, whereas the correct answer and every distractor are referenced to the  $\alpha$ -mixed control.

Query perturbations in these two tasks are not arbitrary. Among perturbations that pass a split-half reliability floor in both contexts and a cross-context response cosine below 0.9, the analysis uses the most divergent: 12 drugs for each cell-line pair and 6 Frangieh knockouts. These two tasks therefore measure retrieval where cross-context divergence is largest rather than on a representative perturbation panel.

Populations contain 400 cells in the controlled task and 200 in the other two, sampled without replacement where the source pool allows it and with replacement otherwise. Within one task setting every query is drawn from a single fixed split of that setting’s cells and differs only in the perturbation or mechanism sampled, in  $\alpha$  and in the resampling seed, so queries within a setting are not independent replicates. The controlled and cross-line tasks use ten resampling seeds and the Frangieh task five.

Hit@1 is reported as an unweighted macro-mean over the seven task–setting combinations, so a 30-query combination counts as much as a 360-query one. Weighting instead by query count gives 0.884 for energy distance and 0.421 for mean cosine, and PCA-space retrieval reaches 0.778 against 0.518 for the corresponding mean-based score.

Hit@1 is the fraction of queries in which no candidate scores strictly higher than the both-mode candidate; its rank is one plus the number of strictly higher-scoring candidates, and the same definition underlies Hit@5, MRR and nDCG@10. That convention would resolve an exact tie in the both-mode candidate’s favour, but no tie arose: across the 1,260 queries and all nine scorers the both-mode candidate is never tied with another candidate at any rank, so every Hit@1 reported here is identical under optimistic, pessimistic and fractional tie handling. The partial-observation protocol below is scored differently: it ranks by a label-blind stable sort and its Hit@1 is the best rank attained by any member of the query’s mechanism or target set, not the rank of a single correct candidate. Ranking a single-mode candidate first counts as a miss, including in the cross-line and Frangieh tasks, where that candidate carries the correct perturbation observed in one of the two contexts. The criterion therefore scores the composition of the retrieved population as well as the mechanism or perturbation it was drawn from. Comparisons between population-level and mean-signature retrieval use the same queries and candidate libraries. Aggregate Hit@1 values are reported as unweighted macro-means over the seven task–setting combinations used in the observed-population analysis; query-weighted averages are reported as a sensitivity analysis so that the largest task does not determine the primary summary.

**Partial-observation protocol.** We evaluated retrieval when only part of each candidate response population was available to the retriever. This protocol separates the information used to construct a ranking from the criterion used to evaluate it. For each query, the population-level and mean-signature methods rank the same candidate library once; the resulting rankings are retained while different evaluation criteria are applied.

Three splits construct the partial observation, all on SciPlex3, and in each the hidden material never enters the candidate library<sup>37</sup>. In *leave-drug-out*, one drug is hidden, the query is that drug’s own treated

population (up to 200 cells) and the library is every other drug in the context. In *leave-MoA-out*, an entire annotated mechanism class is removed from the library and the query is one drug drawn from the removed class, so recovery requires a same-target surrogate among the remaining drugs. In *partial-library*, a fraction of the drugs is treated as measured, the query is built from the pooled responses of the unmeasured drugs, and the retriever selects the next batch of candidates. In all three, query states are obtained by  $k$ -means with  $k = 2$  on the query cells and the minority state is the smaller cluster; candidate populations are capped at 120–150 cells per drug.

The response-matching criterion is an energy-based welfare regret. Each candidate’s welfare is the aggregate over query states  $s$  of its negative energy distance to that state,  $U(d) = \text{agg}_s (-\mathcal{E}(P_d, Q_s))$  with the aggregator taken as the minimum in the worst-case form, and the regret of the selected candidate  $\hat{d}$  is

$$\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0,$$

where  $\mathcal{L}$  is the observed library, so the reference is the best candidate available in the library. Minority-state coverage is reported on the same rankings as the cosine between the candidate’s mean response and the mean response of the query’s minority state, rescaled from  $[-1, 1]$  to  $[0, 1]$ . Regret is zero when the top-ranked candidate is optimal under that objective and increases as the selected candidate becomes less favourable. We report regret reduction relative to mean-signature retrieval, such that a positive value indicates lower regret under this response-matching criterion. This criterion is closely aligned with the population-level information used by the retriever and measures response fidelity, not an independent functional outcome.

**Task-proximal biological evaluation.** The same fixed rankings were next evaluated using mechanism-of-action (MoA) recovery and minority-state coverage. MoA recovery was quantified by normalized discounted cumulative gain (nDCG)<sup>38</sup>, which rewards candidates sharing the query perturbation’s annotated mechanism when they occur near the top of the ranked list. Mechanism recovery is scored on the 600 leave-drug-out queries, that is on the 765 partial-observation queries less the 120 leave-MoA-out and 45 partial-library queries, whose candidate libraries contain no member of the query mechanism class. Within the 600, 21 queries contain no mechanism-class member and score zero for every method; they are retained as ties. Excluding them leaves the median difference at 0.000 and changes the mean from  $-0.024$  to  $-0.025$  and the decreased and increased shares from 40.8% and 35.7% to 42.3% and 37.0%.

Minority-state coverage asks whether the top-ranked candidate better preserves the response of the less represented cellular state in the controlled heterogeneous-population construction. The criterion is applied to the fixed rankings, so it measures how those rankings perform under the biological evaluation.

**Cross-setting projection.** To examine whether the minority-state result extends beyond the controlled partial-observation setting, we evaluated the top-ranked population-level and mean-signature candidates across 239 retrieval task instances spanning natural and constructed settings. These comprised 90 constructed cross-cell-line SciPlex3 tasks, 90 within-cell-line SciPlex3 tasks, 23 Frangieh tasks, 12 CD34+ tasks and 24 tasks using predicted candidate populations. For each task we report the difference in minority-state coverage between the population-level and mean-signature selections.

The 239 tasks form a distribution of task-level differences. The constructed SciPlex3 tasks share data source, preprocessing and candidate structure, and the three SciPlex3 cell lines represent leave-one-cell-

line-out variation within one experiment. We report the continuous difference in coverage.

## External functional evaluation using GDSC2

To test whether transcriptomic retrieval rankings capture information that extends beyond the single-cell expression measurements used to construct them, we evaluated the SciPlex3 rankings against drug-response profiles from the Genomics of Drug Sensitivity in Cancer<sup>28,29</sup>, using the GDSC2 screening set (release 8.5; Data availability). The external criterion was defined as similarity between the functional response profiles of two drugs, matching the retrieval objective of finding a candidate response similar to the query.

SciPlex3 compounds were matched to GDSC2 by normalized drug name, with substring matching used only as a fallback. After removal of one duplicate mapping and restriction to the GDSC2 screening platform, 35 compounds remained. GDSC1 and GDSC2 measurements were analysed separately because they derive from different assay and dose-range settings. For each matched compound, its functional profile was the vector of fitted dose-response AUC values across GDSC2 cell lines. The three cell lines used for the transcriptomic retrieval experiments, A549, K562 and MCF7, were excluded from these profiles, leaving 966 cell lines, so the functional evaluation was performed in biological contexts not used to construct the RNA rankings.

Cell lines differ systematically in their overall sensitivity to drug treatment, which can induce positive correlations between otherwise unrelated compound profiles. We therefore removed this shared cell-line effect before calculating drug-drug similarity. For cell line  $l$  and compound  $d$ , the centred response was

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where  $\mathcal{D}_l$  contains all GDSC2 compounds measured in cell line  $l$ , rather than only the 35 compounds overlapping SciPlex3. Functional similarity between query drug  $q$  and candidate drug  $c$  was then defined as

$$F(q, c) = \rho_s \left( \widetilde{\text{AUC}}_{q,\bullet}, \widetilde{\text{AUC}}_{c,\bullet} \right),$$

where  $\rho_s$  denotes Spearman correlation across cell lines shared by the two compounds. Drug pairs measured in fewer than 100 common cell lines were excluded. This yielded 591 usable drug pairs, with a median of 909 shared cell lines per pair. The corresponding uncentred functional similarity was retained as a sensitivity analysis.

For each SciPlex3 cell line, retrieval was evaluated in a leave-one-drug-out design. The 10  $\mu\text{M}$  response population of the held-out compound served as the query, and the remaining matched compounds formed the candidate library, giving 34–35 candidates per query. Each RNA-based retrieval rule produced a candidate ranking without access to GDSC2 data. Agreement between that ranking and the external functional ordering  $F(q, c)$  was quantified by Spearman correlation across candidates.

The primary RNA-derived rankings were energy-distance population retrieval and control-referenced mean-cosine retrieval. To test whether apparent functional agreement required cell-to-cell distributional information, we additionally used a response-magnitude matching score,

$$M(q, c) = - \left| \|\vec{d}_q\| - \|\vec{d}_c\| \right|,$$

which prefers candidates whose overall transcriptional response magnitude is similar to that of the query. A query-independent magnitude score  $\|\vec{d}_c\|$  was retained as a diagnostic control. An uncorrected

mean-cosine score based on treated means without matched-control subtraction and a potency-matching diagnostic were analysed were included as sensitivity controls.

To estimate the component of the energy-based association not explained by response magnitude, we computed a partial rank correlation with the magnitude-matching channel removed. Retrieval scores and functional similarities were rank-transformed, each was residualized with respect to  $M(q, c)$ , and the correlation between the two residuals was reported as the magnitude-adjusted association.

Per-query Spearman correlations are summarized by their median within each SciPlex3 cell line. The three cell lines provide the higher-level analysis unit; query-level paired tests are descriptive analyses where indicated.

The comparison between energy retrieval and the response-magnitude scalar is pre-specified as a pooled test across the three SciPlex3 cell lines; per-line ratios are given in Supplementary Methods.

### Protein-based evaluator-shape experiment

Frangieh Perturb-CITE-seq<sup>30</sup> provides matched RNA and surface-protein measurements from the same cells, allowing the statistical form of an external evaluator to be changed while holding its biological measurement fixed. RNA and protein matrices were joined by cell barcode before analysis; row position was not used because the released modality-specific matrices differ in cell number and ordering. Of the 24 measured antibodies, four isotype controls were removed after normalization, leaving 20 surface-protein features. Antibody-derived tag counts<sup>39</sup> were centred-log-ratio normalized within each cell,  $y_{ij} = \ln(1 + x_{ij}) - m^{-1} \sum_k \ln(1 + x_{ik})$  for cell  $i$  and antibody  $j$ , with a pseudocount of one, a natural logarithm and the per-cell mean taken over all  $m = 24$  measured antibodies including zero counts and the four isotype controls; normalization therefore precedes isotype removal. This is the centred log-ratio of the  $(1 + x)$  vector and differs from implementations that exclude zero entries from the geometric mean.

All analyses were performed separately within the Control, IFN $\gamma$  and Co-culture conditions to avoid mixing immune context with perturbation identity. Knockouts represented by at least 60 barcode-matched cells were retained, and populations were subsampled to at most 250 cells using a fixed random seed.

**Protein-based evaluators.** For knockout  $k$  in immune condition  $c$ , let  $\bar{d}_{k,c}^{\text{prot}}$  denote its mean control-referenced protein response and let  $D_{k,c}^{\text{prot}}$  denote the corresponding set of per-cell protein responses. The mean-based evaluator was defined as

$$E_{\text{mean}}(q, d) = \cos \left( \bar{d}_{q,c}^{\text{prot}}, \bar{d}_{d,c}^{\text{prot}} \right).$$

The population-based evaluator was defined from the same cells and protein measurements as

$$E_{\text{pop}}(q, d) = -\mathcal{E} \left( D_{q,c}^{\text{prot}}, D_{d,c}^{\text{prot}} \right),$$

where  $\mathcal{E}$  is the energy distance. The negative sign converts distance to similarity so that larger values indicate a closer match under both evaluators.

For both definitions, protein responses were referenced to the matched control population from the same immune condition. Thus,

$$\bar{d}_{k,c}^{\text{prot}} = \text{mean} \left( X_{k,c}^{\text{prot}} \right) - \mu_{0,c}^{\text{prot}},$$

625 and

$$D_{k,c}^{\text{prot}} = \left\{ x_{i,k,c}^{\text{prot}} - \mu_{0,c}^{\text{prot}} \right\}_{i=1}^{n_{k,c}}.$$

626 Using the same matched control for the two evaluator forms isolates the effect of summarizing protein  
627 responses by their mean or retaining them as a cellular population.

628 **Fixed RNA-derived rankings.** The RNA-derived rankings were computed once and evaluated against  
629 both protein criteria. The primary rankings were energy-distance population retrieval and control-  
630 referenced mean-cosine retrieval. The query populations, candidate libraries and RNA-derived candi-  
631 date scores were identical under the two protein evaluators, and no protein measurement was available  
632 to any retrieval method. The experiment therefore changes only the statistical representation used to  
633 judge the same RNA-derived ranking.

634 **Response-magnitude control.** Because a population distance can also encode differences in overall re-  
635 sponse magnitude, we included a query-dependent scalar control that uses no cell-to-cell distributional  
636 information. For query  $q$  and candidate  $d$ , magnitude matching was defined as

$$M(q, d) = - \left| \left| \bar{d}_{q,c}^{\text{RNA}} \right| \right| - \left| \left| \bar{d}_{d,c}^{\text{RNA}} \right| \right|.$$

637 This ranking favours candidates with a transcriptional response magnitude similar to that of the query  
638 while discarding all within-population structure. The same magnitude ranking was evaluated against  
639 both protein criteria.

640 For each query, agreement between an RNA-derived ranking and a protein-based evaluator was  
641 quantified by Spearman correlation across that query’s candidate knockouts. The statistic is therefore a  
642 rank correlation over candidates, computed once per query, and never a correlation over pooled query-  
643 candidate pairs. The headline value for each ranking and each evaluator form is the median of those  
644 per-query correlations over all 659 queries. The same medians were also taken within each of the three  
645 immune conditions, which are reported separately and treated as the higher-level analysis unit; queries  
646 within a condition share candidate structure and were not interpreted as independent biological repli-  
647 cates.

## 648 Natural heterogeneity and large-scale validation

649 We next examined the information requirements for population-level retrieval in datasets in which cellu-  
650 lar heterogeneity arose within biological samples. Patient-derived glioblastoma was used to test the re-  
651 quirements in naturally heterogeneous tissue, and Tahoe-100M was used to assess the same quantities at  
652 substantially larger scale. Tahoe-100M also provided the substrate for the source-to- target intervention-  
653 identification benchmark described below.

654 **Patient-derived glioblastoma.** The ZhaoSims2021 dataset<sup>16</sup> contains acute-slice cultures and surgi-  
655 cal biopsies from ten glioblastoma patients exposed to six compounds: Ana-12, Ispenisib, RO4929097,  
656 Tazemetostat, etoposide and panobinostat. Expression counts were normalized to  $10^4$  per cell, log1p-  
657 transformed and reduced to 2,000 highly variable genes. Only one patient received more than two com-  
658 pounds. We therefore measure state-specific response, recoverability and the extent to which population-  
659 level similarity differs from compartment-specific response ordering.

660 **Cellular-compartment assignment.** Cells were assigned to malignant and myeloid compartments using canonical marker scores; ambiguous cells were excluded. Drug responses were defined relative to patient- and compartment-matched controls. Assignment criteria, retained-cell counts and threshold-sensitivity analyses are reported in Supplementary Note 3.

664 **Differential response in natural tissue.** State-specific perturbation responses were measured between malignant and myeloid compartments. For patient  $p$ , drug  $d$  and compartment  $k$ , the response vector was

$$r_{p,d,k} = \text{mean} \left( X_{p,d,k}^{\text{treated}} \right) - \text{mean} \left( X_{p,k}^{\text{control}} \right).$$

667 Only patient–drug combinations with at least 50 treated cells in both compartments were included. Controls were matched within both patient and compartment. Differential response was summarized as

$$R_{p,d} = \cos \left( r_{p,d,\text{malignant}}, r_{p,d,\text{myeloid}} \right).$$

669 This construction removes pre-existing differences between patients and cellular compartments before comparing the directions of their drug-induced responses.

671 **Recoverability in natural tissue.** Response-identity recoverability was assessed from within-patient drug-pair splits by comparing unsupervised partitions with a supervised ceiling. Patient-level bootstrap and method-pair sensitivity analyses are reported in Supplementary Note 3.

674 **Proxy for downstream decision relevance.** The glioblastoma cohort contains insufficient drug coverage for direct retrieval. We quantified whether whole-population drug similarity preserves the ordering observed within a specific cellular compartment.

677 For each pair of drugs measured in the same patient, we first computed the cosine similarity between their population-mean response signatures and the cosine similarity between their malignant-compartment response vectors. Agreement between these two similarity structures was summarized by Spearman correlation across drug pairs.

681 That whole-population comparison gives a Spearman correlation of +0.878. Because the population-mean signature is the equal-weight average of the two compartment responses, the malignant compartment contributes half of it. We also used a disjoint comparison. For drug pair  $(a, b)$ , myeloid-response similarity was defined as

$$S_{a,b}^{\text{myeloid}} = \cos \left( r_{a,\text{myeloid}}, r_{b,\text{myeloid}} \right),$$

685 and malignant-response similarity as

$$S_{a,b}^{\text{malignant}} = \cos \left( r_{a,\text{malignant}}, r_{b,\text{malignant}} \right).$$

686 We then computed

$$\rho_{\text{disjoint}} = \rho_S \left( S_{a,b}^{\text{myeloid}}, S_{a,b}^{\text{malignant}} \right),$$

687 using only drug pairs available within the same patient. The two sides contain no shared cells and each

688 compartment is referenced to its own matched control. This statistic measures the candidate- ordering  
689 headroom available to population structure.

690 **Large-scale validation in Tahoe-100M.** Tahoe-100M plate 3 was used to test the same requirements at  
691 larger scale. We evaluated two independently defined states, released cell-cycle labels and a control-  
692 derived  $k$ -means partition, and quantified state-specific response alignment, response-identity recover-  
693 ability and agreement of drug ordering across disjoint states. Dataset filtering and all state-definition  
694 details are reported in Supplementary Note 3.

695 For each eligible drug-cell-line condition, the response of state  $k$  was defined relative to its own  
696 state-matched controls,

$$r_{d,c,k} = \text{mean} \left( X_{d,c,k}^{\text{treated}} \right) - \text{mean} \left( X_{c,k}^{\text{control}} \right).$$

697 Differential response was measured as the cosine between the two state-specific response vectors,

$$R_{d,c} = \cos \left( r_{d,c,0}, r_{d,c,1} \right).$$

698 A condition was included only when at least 50 cells were available in each required treated and control  
699 state.

700 Finally, a disjoint state-ordering statistic was used as a large-scale proxy for downstream decision  
701 relevance. Within each cell line, cosine similarity was computed for every eligible pair of drug responses  
702 separately in the two non-overlapping cellular states. Spearman correlation between the two resulting  
703 drug-drug similarity matrices was then calculated,

$$\rho_c = \rho_s \left( S_{a,b}^{(0)}, S_{a,b}^{(1)} \right).$$

704 High correlation indicates that the two cellular states preserve similar relative ordering among drug  
705 responses; lower correlation leaves greater scope for state-specific information to change candidate pref-  
706 erence. This statistic is a proxy for decision relevance.

707 These intrinsic-state diagnostics are reported separately from the intervention-identification bench-  
708 mark below, which uses the same plate. Detailed recoverability methods are in Supplementary Note 3.

### 709 **Source-to-target intervention identification on Tahoe-100M**

710 The retrieval experiments above rank candidates whose responses in the query context are observed.  
711 The intervention task asks which compound carries a starting population into a desired one when the  
712 candidate’s effect in that context is unknown. The desired population is a real treated population gen-  
713 erated by one compound of the library, and the outcome is the rank assigned to that compound. We  
714 defined this task on Tahoe-100M plate 3; it measures intervention identification rather than therapeutic  
715 utility.

716 **Task definition and eligibility.** A context is one cell line. For a held-out context  $c$  and a query drug  $q$ ,  
717 the retriever receives the vehicle (DMSO) cells of  $c$  and an observed treated population  $X_{q,c}$ , and ranks  
718 all 92 candidate compounds; it never sees  $q$ . A condition entered the grid if it had at least 100 treated and  
719 200 vehicle cells, which admitted 3,992 eligible queries over 44 usable cell lines from the 4,600 conditions  
720 present on the plate. Each query was evaluated at five fixed seeds (13, 29, 47, 71, 101), giving 19,960  
721 query-seed pairs. Source and target populations were drawn without replacement at up to 200 cells

each, all available cells being taken where a condition held fewer, and responses were referred to the source mean,  $P = \{x_i - \bar{x}_{\text{source}}\}$ .

**Oracle arm.** In the oracle arm every candidate’s response in the held-out context is taken from the observed data, isolating the retrieval step and defining an information ceiling for prediction. Cell disjointness is enforced through the library preparation: the target population is drawn from one sublibrary half and every candidate bank from the other, at up to 100 cells per candidate, all cells in that half being taken where a candidate has fewer, without replacement, so the generating drug is drawn from the same half as its 91 competitors and is not penalized by a technical shift.

**Prediction arm.** In the prediction arm no treated cell of the held-out context enters any fit or any ranker. Candidate responses are produced by a forward predictor fitted on the other 43 cell lines. Six predictors were evaluated. Three are additive: they emit one signature per drug and apply it identically to every source cell, so the candidate population is the vehicle population rigidly translated. These are an average effect  $\delta_d$  pooled over training contexts; a nearest-context transfer taking  $\delta_d$  from the training context whose vehicle mean is closest in cosine; and a linear-latent variant truncating the average effect to the principal components carrying 80% of training variance. Three preserve cell-specific response structure. The *state-conditioned* predictor estimates the same average effect separately for each of the three cell-cycle states released with the plate, G1, S and G2M, and applies it to each source cell according to its own state, which costs one extra grouping and no new hyperparameter. Each per-state effect is averaged over the training contexts in which that state holds at least 20 cells; groups below the floor are omitted, and 359 of the 12,275 cell-line, drug and state groups fall below it, 294 of them in S phase. A drug left with no usable estimate for a state would keep its pooled effect there and be additive for that state alone; this did not occur for any of the  $44 \times 92 \times 3$  signature slots. The predictor is non-additive over its whole output. The *Gaussian optimal-transport* map is a Bures map fitted in a 50-component training principal-component basis, from the pooled within-context vehicle covariance to the drug’s pooled within-condition treated covariance, with the mean shifted by the average latent effect and applied to the held-out vehicle cells as a latent edit so that the part of each cell outside the subspace is carried through; every cloud is centred on its own condition mean before pooling, so a covariance here is within-condition cell scatter. *Nearest context, cells* transfers the neighbouring context’s actual treated cells for the drug, shifted by the difference of the two vehicle means; it is the only low-cost predictor in this set that can carry a non-additive response shape. Each principal-component basis is fitted on up to 50,000 cells sampled from the training contexts, and each context–drug fit uses up to 200 cells. The additive three generate no state-specific differential response, an algebraic property of the model class.

**Scoring and evaluation.** Candidates were scored by the three rules of Fig. 1a: a direction-only mean cosine, a magnitude-aware mean  $L_2$ , and the population route, primarily the unbiased U-statistic energy distance, with MMD and sliced Wasserstein as secondary variants. The secondary scores use settings specific to this task: sliced Wasserstein over 200 random projections at 100 quantiles, and a single RBF bandwidth fitted once by the median heuristic on the candidate library and held fixed across queries. Mean reciprocal rank of the true intervention is the primary outcome; Hit@1, Hit@5 and Hit@10 are secondary. Intervals are 2,000-fold bootstrap intervals resampled over cell-line clusters, which are the independent units of this design. A decision is counted as corrected when the population route places the true drug first and the reference does not, and as broken in the reverse case.

## 763 Headroom, response detectability and the conditional bottleneck analysis

764 **Headroom and its null.** Headroom is defined per query as  $H = 1 - \text{RR}_{\text{ref}}$ , the room left by the  
 765 direction-only mean route, and the gain is  $G = \text{RR}_{\text{pop}} - \text{RR}_{\text{ref}}$  on the same query. A Spearman cor-  
 766 relation between  $H$  and  $G$  includes the shared reference rank and the reciprocal-rank ceiling. Within  
 767 each cell line, the population route’s reciprocal ranks are randomly permuted across queries, preserving  
 768 both marginal distributions, the  $\text{RR} \leq 1$  ceiling and the cell-line clustering;  $G$  is then recomputed against  
 769 the unpermuted reference and the correlation retaken. Two hundred permutations at a fixed seed give  
 770 the interval reported in Fig. 5e.

771 **Response detectability.** For each held-out cell line  $c$  and query drug  $q$ , equal numbers of cells treated  
 772 with  $q$  and vehicle cells of  $c$  form a balanced two-class problem, at 300 per arm or at the size of the  
 773 smaller pool where that is below 300, sampled at a fixed seed. Chance accuracy is 0.5. The supervised  
 774 ceiling  $A_{\text{sup}}$  is the out-of-fold accuracy of a five-fold cross-validated logistic probe ( $C = 1$ ) on those cells.  
 775 The unsupervised value  $A_{\text{unsup}}$  is  $k$ -means with  $k = 2$  on the same cells, scored by best-permutation  
 776 accuracy. The labels are external, defined by the source well rather than by an expression-derived cell  
 777 state. The supervised arm is a ceiling on the information present in the representation.

778 **Conditional bottleneck regression.** Regressing the gain  $G$  directly on candidate explanators includes  
 779 the same shared reference rank as the headroom correlation. The specification reported in Fig. 5h there-  
 780 fore takes the population route’s own reciprocal rank as the outcome and includes the mean route’s  
 781 reciprocal rank as a covariate, so that a coefficient describes how the explainer relates to population-  
 782 route performance at a given mean-route rank. The explanators are the interaction share, its cross-fitted  
 783 reproducibility  $R_{\text{int}} = \cos(I_A, I_B)$ , response detectability  $A_{\text{sup}}$ , and, in a separate model reported for con-  
 784 trast, the naive response cosine  $1 - \cos(r_1, r_2)$ . Estimation is ordinary least squares with cluster-robust  
 785 standard errors at the cell line. All predictors are standardised to zero mean and unit variance so that  
 786 coefficients are comparable across explanators. Queries missing any explainer are dropped listwise;  
 787 the reported  $n$  is 3,618.

## 788 Cross-fitted drug-by-state interaction

789 Testing whether a drug’s effect depends on cell state requires a statistic that measures interaction. Writ-  
 790 ing  $r_s$  for the state- $s$  response, the difference between mean treated and mean vehicle cells of that state,  
 791 with states 1 and 2 the G1 and G2M cell-cycle calls released with the plate, the quantity of interest is the  
 792 interaction vector  $I = r_1 - r_2$ . Its squared norm is sensitive to sampling noise: with finite arms, noise  
 793 alone makes  $\|I\|$  large, and the same mechanism inflates  $1 - \cos(r_1, r_2)$  for the weakest drugs.

794 Each of the four arms is therefore split into disjoint halves  $A$  and  $B$ , the interaction is estimated once  
 795 in each, and the two estimates are multiplied:

$$S_{\text{int}} = \frac{\langle I_A, I_B \rangle}{p}, \quad S_{\text{main}} = \frac{\langle M_A, M_B \rangle}{p}, \quad M = \frac{1}{2}(r_1 + r_2),$$

796 with  $p$  the number of genes. Under the additive null  $E[I] = 0$  and the two halves are independent,  
 797 so  $E[S_{\text{int}}] = 0$  exactly: noise makes  $I_A$  and  $I_B$  point in unrelated directions and their inner product  
 798 averages to zero, whereas a real state-dependent effect survives because both halves estimate the same  
 799 non-zero  $I$ . The vehicle arms are split as well; sharing one vehicle mean between halves would place the  
 800 same control noise in  $I_A$  and  $I_B$  and correlate them under the null. Splits are library-preparation-disjoint

801 wherever a condition spans at least two sublibraries, and a plain random half of the cells otherwise.  
802 Arms were capped at 400 cells and required at least 50, and ten A/B splits were averaged per query.

803 Two derived quantities are used. The *interaction share*,  $S_{\text{int}} / (S_{\text{int}} + S_{\text{main}})$ , is scale-free and uses cross-  
804 fitted numerator and denominator.  $R_{\text{int}} = \cos(I_A, I_B)$  asks whether the interaction reproduces across  
805 halves. As a positive control, synthetic conditions were built from the plate's own cells by taking the G1  
806 cells from one drug and the G2M cells from another, so interaction is present by construction in material  
807 with the same states and arm sizes as the queries.

808 The cosine statistic  $1 - \cos(r_1, r_2)$  is retained as a descriptive sensitivity analysis and as a comparator  
809 in Fig. 5g. It is not used as a differential-response measure anywhere that a conclusion depends on  
810 it. Where earlier analyses in this study used the induced-response cosine to characterize predictors or  
811 tissue, that use is descriptive alignment between two response directions and is labelled as such.

## 812 Statistical analysis

813 Statistical units were defined by the experimental design: cells estimate a perturbation-response popu-  
814 lation, and retrieval performance is summarized at the designated cluster or task level<sup>40</sup>.

815 The partial-observation arm uses the query drug as the primary inferential cluster. Of the 765 queries,  
816 720 hold out one of 143 drugs and the remaining 45 pool the unmeasured drugs of the partial-library split  
817 into a single batch query; those pooled queries remain in descriptive summaries but are excluded from  
818 formal drug-cluster inference. The 600 leave-drug-out queries on which mechanism recovery is scored  
819 come from 130 drugs. Each drug is scored across three shared cell lines, several library fractions and  
820 up to five random seeds, giving 417 distinct drug-by-cell-line contexts in the individual-drug universe.  
821 Primary 95% intervals for this arm use 10,000 percentile bootstrap resamples of query-drug clusters, re-  
822 taining all rows belonging to each sampled drug. For paired comparisons, repeated observations are first  
823 collapsed to one median effect per drug and a two-sided drug-level Wilcoxon signed-rank test is used  
824 when reported. For the diagnostic recommended-versus-no-call comparison, inference is summarized  
825 by the cluster-bootstrap interval because drugs can contribute to both groups.

826 Paired comparisons between retrieval methods outside the clustered partial-observation analysis  
827 used two-sided Wilcoxon signed-rank tests; independent groups used Mann–Whitney  $U$  tests; mono-  
828 tonic associations used Spearman rank correlation. Across pre-specified divergence strata, drug-level  $P$   
829 values were adjusted by the Benjamini–Hochberg procedure<sup>41</sup>. Effect sizes accompany all tests. Power  
830 calculations for the relevant stratum are reported in Supplementary Note 1. Mechanism-of-action analy-  
831 ses used the 600 queries whose mechanism class remained represented in the candidate library; queries  
832 without a defined target were excluded.

833 For the GDSC2 functional analysis, Spearman correlations were computed per query across its candi-  
834 date drugs and summarized within each SciPlex3 cell line, which is the analysis unit; query-level paired  
835 tests are reported as nominal. The protein evaluator-shape experiment was summarized within the Con-  
836 trol, IFN $\gamma$  and Co-culture conditions, its three higher-level units. For HIR-Bench, cross-validation was  
837 grouped by the label-determining ( $\alpha$ , conflict) pair, giving 28 effective parameter cells, and performance  
838 was summarized by pooled out-of-fold AUC. For the patient-derived glioblastoma analysis, uncertainty  
839 in the recoverability gap was estimated by cluster bootstrap at the patient level with 5,000 resamples, dif-  
840 ferences being taken within each drug-pair split before aggregation; four patients contributed evaluable  
841 splits. Intervention-retrieval intervals are 2,000-fold bootstrap resamples over cell-line clusters. Tahoe-  
842 100M summaries are taken across eligible drug–cell-line conditions or within-cell-line drug pairs accord-  
843 ing to the statistic, and both cellular-state definitions, released cell-cycle state and the control-derived

844 *k*-means partition, were analysed in parallel.

845 Analyses were implemented in Python. Single-cell data handling used Scanpy<sup>42</sup>; the *k*-means,  
846 Gaussian-mixture, principal-component and logistic-regression procedures named above are the  
847 scikit-learn implementations<sup>43</sup>.

## 848 Data availability

849 All datasets analysed in this study are publicly available.

850 SciPlex3, Frangieh Perturb-CITE-seq and ZhaoSims2021 were obtained in harmonized form  
851 from scPerturb<sup>13</sup>, Zenodo record 13350497 ([doi:10.5281/zenodo.13350497](https://doi.org/10.5281/zenodo.13350497), release 1.4). The exact  
852 harmonized files used in the analyses were SrivatsanTrapnell2020\_sciplex3.h5ad for SciPlex3,  
853 FrangiehIzar2021\_RNA.h5ad and FrangiehIzar2021\_protein.h5ad for Frangieh Perturb-CITE-seq, and  
854 ZhaoSims2021.h5ad for the patient-derived glioblastoma analysis. The primary SciPlex3 data are avail-  
855 able from the Gene Expression Omnibus under accession GSE139944<sup>14</sup>; the original Frangieh and  
856 ZhaoSims2021 datasets are described in the corresponding publications<sup>16,30</sup>. The CD34+ haematopoietic  
857 progenitor dataset is available from the Gene Expression Omnibus under accession GSE306429<sup>32</sup>.

858 Tahoe-100M<sup>15</sup> was released by Vevo Therapeutics and the Arc Institute under CC0 and is distributed  
859 through the Arc Virtual Cell Atlas and Hugging Face (tahoebio/Tahoe-100M). This study uses plate 3  
860 only, together with the released obs\_metadata and sample\_metadata tables. Analyses use the released  
861 preprocessed expression matrix comprising 4,158,278 cells and 2,304 highly variable genes in log1p-  
862 transformed space.

863 The external functional evaluation used GDSC2 fitted dose-response data from the Genomics of  
864 Drug Sensitivity in Cancer, release 8.5 (GDSC2\_fitted\_dose\_response\_270ct23.xlsx), available from  
865 <https://www.cancerrxgene.org>. The compound-matching table used to align the SciPlex3 and GDSC2  
866 drug panels (drug\_match\_table.csv) is provided with the analysis code.

867 Processed analysis inputs and the scripts used to construct them from the source datasets are pro-  
868 vided in the accompanying repository. Source-data tables underlying each figure panel are provided  
869 with the corresponding analysis outputs.

## 870 Code availability

871 All code used for data processing, retrieval, evaluation and statistical analysis is available at <https://github.com/Boom5426/PopRetrieve>.  
872

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## 875 Author contributions

## 876 Competing interests

877 The authors declare no competing interests.

## 878 Additional information

879 Supplementary Information is available for this paper: Supplementary Methods, Supplementary Notes  
880 1–4 and Supplementary Tables 1–4.

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